

CoalGameRec: validation-guided interaction attribution for graph recommendation — a frozen LightGCN study of LOO versus bounded Shapley

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Abstract

Graph-based recommenders rank accurately, but it remains unclear whether interaction-level attribution improves reranking beyond simple reweighting. We present CoalGameRec, a validation-guided attribution framework for a frozen LightGCN backbone. Selected historical interactions are treated as players, and coalition utility is a validation-only pairwise log-sigmoid score. We compare a bounded antithetic Monte Carlo Shapley estimator ($k = 24$, $M = 64$) with the leave-one-out (LOO) marginal $v(N) - v(N \setminus \{j\})$ under temporal leave-one-out, full-catalog ranking, five seeds (42–46), and paired user bootstrap inference ($B = 2000$) on MovieLens-1M and a temporal Amazon-Book split. LOO yields NDCG@20 0.04976 ± 0.00041 and 0.03237 ± 0.00077 , compared with 0.04601 ± 0.00030 and 0.02978 ± 0.00078 for uniform weighting (+8.1% and +8.7%); within the Holm-adjusted primary family, Shapley beats uniform with $p < 0.0005$ on both datasets; Shapley also beats heuristic controls but is not consistently ahead of LOO — on MovieLens-1M HitRate@20 Shapley is higher (0.12555 vs 0.12519, n.s. $p = 0.575$). Attribution time is $13.0\times$ (Amazon) to $15.7\times$ (ML-1M) lower for LOO; NDCG gain per attribution hour is $16.1\times$ – $18.3\times$ higher. A confirmatory re-execution of the frozen protocol with matched validation-informed non-game controls (valid-similarity reweighting and a validation-tuned linear reranker) shows that LOO also beats both matched controls on NDCG@20 on both datasets (Holm $p \leq 0.004$), closing the validation-access asymmetry. Results support validation-guided marginal attribution as a deployable reranking signal, but they do not establish that full Shapley averaging improves ranking or that attributions are faithful explanations.

Keywords: recommender systems, graph neural networks, LightGCN, explainable AI, co-operative game theory, Shapley value, attribution, ranking, cost-effectiveness

1 Introduction

Graph-based recommender systems model user preferences through observed interactions [1, 2, 3], social links, knowledge graphs, or higher-order relational structures. In top-N recommendation, graph neural methods such as NGCF and LightGCN [4, 5, 6] propagate user and item representations over a bipartite interaction graph and often outperform matrix-factorization baselines [5, 6]. Their success creates a familiar problem: the model can rank items effectively, but it is rarely clear which historical interactions are responsible for a recommendation.

Explainable and causal recommendation [7, 8] addresses this gap by producing evidence for a recommendation in terms of features, items, paths, neighbors, providers, or user actions. Cooperative-game attribution is a particularly attractive candidate because it defines a value function over coalitions of players and allocates the resulting utility through solution concepts such as the Shapley value, Banzhaf value, Myerson value, or interaction indices. For recommendation, the players may be historical user-item interactions, graph edges, item features, signal sources, providers, or training data points. The value may be ranking quality, score margin, exposure, diversity, or another task-specific objective.

Despite this appeal, cooperative-game attribution in graph recommendation is not yet a settled methodology. Published methods differ in what counts as a player, which coalition semantics are used, whether the model is retrained or masked, whether the value function is smooth or discontinuous, and how attribution is used after it is computed. A Shapley value over one game may be theoretically well-defined but scientifically uninformative if the players, value function, or intervention do not correspond to the explanation question.

This paper has two aims. First, we formalize a five-principle design framework and a five-axis taxonomy for cooperative-game attribution, making explicit when game-theoretic language is substantive and when it collapses to similarity reweighting. Second, we test a central empirical claim: does validation-guided interaction attribution, instantiated as a frozen-graph game with a smooth pairwise utility, improve held-out ranking and at what cost, and does full Shapley averaging beat its single-context LOO marginal? We instantiate the framework with LightGCN and report a five-seed study (protocol frozen 2026-07-15, Supplementary File S1) that establishes a cost-effectiveness frontier against matched heuristic baselines.

The empirical study is deliberately discriminative rather than promotional. We test whether any validation-guided attribution beats heuristic reweighting, and then whether Shapley’s coalition averaging adds value beyond the cheapest game-theoretic baseline: the leave-one-out marginal $v(N) - v(N \setminus \{j\})$. If LOO matches Shapley, the practical gain is from validation-guided marginal signal, not from averaging over many contexts — with direct implications for deployment cost.

1.1 Contributions

The paper makes five contributions.

1. **CoalGameRec**, a validation-guided interaction-attribution framework with two instantiations (bounded Shapley and its efficient LOO marginal) whose LOO instantia-

tion beats every matched non-game control (uniform, additive-pref, attention, popularity) — and, in the confirmatory study C1, also both matched *validation-informed* non-game controls (valid-sim, valid-linear) on NDCG@20 on both datasets — and matches or beats bounded Shapley on NDCG@20 and Coverage@20 (Tables 3, 7, 8).

2. A five-principle design framework (player, coalition semantics, smooth validation value, model-native intervention, mandatory marginal baseline) that makes game-theoretic claims falsifiable and separates genuine attribution from additive leakage $\phi(v_{pref}) = \phi(v) + \lambda \text{sim}$.
3. A five-axis taxonomy (player $\times$ value $\times$ solution $\times$ role $\times$ structure) predicting when averaging over contexts helps and when it collapses to similarity.
4. A reproducible, leakage-free LightGCN protocol (temporal LOO splits, full-catalog ranking, 5 seeds 42–46, $B = 2000$ paired user bootstrap, $M = 64$, $k = 24$ stratified players, native embedding intervention) that isolates attribution signal from training leakage. The primary inferential family is defined explicitly: per dataset, $F = 8$ Holm-adjusted contrasts, Shapley-MC vs. {uniform, additive-pref, attention, LOO} $\times$ {NDCG@20, HitRate@20}; every reported contrast maps one-to-one to a released artifact file.
5. A cost–accuracy frontier and deployment rule: the LOO instantiation of CoalGameRec gives the best NDCG and Coverage at 13.0–15.7 $\times$ lower wall-clock attribution time than Shapley and 16.1–18.3 $\times$ higher NDCG gain per attribution hour; a hybrid Shapley+LOO fusion variant is released in code as a candidate extension but is not evaluated in the frozen protocol.

2 Background and preliminaries

2.1 Graph-based top-N recommendation

Let $\mathcal{U}$ be the set of users and $\mathcal{I}$ the set of items. In implicit-feedback top-N recommendation, the observed training graph contains positive user-item interactions. A graph recommender learns user and item representations and produces a score $s(u, i)$ for each candidate item. Evaluation ranks all unseen eligible items for each user and measures whether the held-out item appears near the top.

This case study uses LightGCN as a graph recommender backbone [6]. Let $\mathbf{E}^{(0)} \in \mathbb{R}^{(|\mathcal{U}|+|\mathcal{I}|) \times d}$ be the layer-0 embeddings. With normalized adjacency $\tilde{\mathbf{A}} = \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$ of the training graph, LightGCN propagates

$$\mathbf{E}^{(l+1)} = \tilde{\mathbf{A}} \mathbf{E}^{(l)}, \quad l = 0, \dots, L-1, \quad (1)$$

and reads out $\mathbf{e}_u = \frac{1}{L+1} \sum_{l=0}^L \mathbf{e}_u^{(l)}$, $\mathbf{e}_i$ likewise, scoring $s(u, i) = \langle \mathbf{e}_u, \mathbf{e}_i \rangle$. Training uses Bayesian Personalized Ranking [9]:

$$\mathcal{L}_{\text{BPR}} = - \sum_{(u, i, j) \in \mathcal{D}} \log \sigma(s(u, i) - s(u, j)) + \lambda_{\text{reg}} \|\Theta\|_2^2, \quad (2)$$

where i is a positive item, j a sampled negative, σ the sigmoid. In this paper LightGCN is the sole backbone (hypergraph variants are future work).

2.2 Cooperative games and Shapley attribution

A cooperative game is defined by a player set N and a characteristic function $v : 2^N \rightarrow \mathbb{R}$ [10]. The SHAP framework [11] popularized Shapley for ML, and Data Shapley [12] priced training points and efficient variants [13]; ShaRP explains rankings [14]; we adapt the same axioms to interaction reranking. The Shapley value allocates the value of the grand coalition to players by averaging each player’s marginal contribution over all coalition contexts. For player p ,

$$\phi_p(v) = \sum_{S \subseteq N \setminus \{p\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{p\}) - v(S)]. \quad (3)$$

In recommendation, N may be a user’s historical interactions. The value function may measure validation ranking utility when only a subset of those interactions is active. The Shapley value then assigns credit to historical interactions according to their expected contribution across contexts.

2.3 Why Shapley may fail to add ranking value

The Shapley value is not a guarantee of better recommendations. It is an allocation rule conditional on a chosen game. If the value function contains an additive similarity term, Shapley linearity decomposes that term into the same per-interaction similarity scores used by non-game baselines. If the value function uses hard top-20 NDCG with one validation item, many coalitions may receive zero utility, producing sparse marginal signals. If the intervention uses an external similarity kernel that is misaligned with the graph backbone, the attribution signal may be distorted before it affects ranking.

These issues motivated the case-study design used here. The primary game removes the additive preference term, uses a smooth validation-only pairwise utility, selects a stratified bounded player set, and applies attribution through the backbone-native item embedding space.

3 A design framework for cooperative-game recommendation attribution

The central contribution of this article is not only a literature map. It is a design framework for deciding when a cooperative-game explanation is meaningful in graph recommendation. The framework is motivated by a simple observation: a Shapley value is an answer to a specific game, not a universal explanation. To evaluate any cooperative-game explanation, the analyst must specify five linked components: the player set, the coalition semantics, the value function, the solution concept, and the intervention or explanation use.

3.1 Principle 1: the player must match the explanation question

A method should not be evaluated only by whether it uses Shapley values. This concern is consistent with critiques in graph-XAI surveys and taxonomies, which emphasize that an explanation method must specify the object being explained and the evidence used to validate the explanation [15]. It should be evaluated by whether its players correspond to the question being asked. If a user asks why a book was recommended, historical interactions may be the relevant players. If a platform asks whether providers are treated fairly, providers may be the relevant players. If a model developer asks whether a knowledge-graph path influenced a score, graph edges or path components may be the relevant players. This principle prevents the common error of citing feature-attribution results as evidence for interaction-level recommendation explanations.

3.2 Principle 2: the value function must be smooth enough to attribute

A discontinuous value function can make attribution unstable. Hard top-K metrics are suitable final evaluation metrics, but they are often poor coalition utilities. A coalition that improves a validation item from rank 200 to rank 30 receives no credit under HitRate@20, even though it improved the model. For this reason, the empirical case study uses a validation-only pairwise log-sigmoid value for attribution and reserves HitRate@20 and NDCG@20 for final test evaluation. This follows the general principle that attribution utilities should be sensitive to the local changes that the attribution method is expected to detect.

3.3 Principle 3: additive terms should not be hidden inside Shapley claims

If the value function contains an additive term, Shapley linearity passes that term directly into the attribution. For example, if

$$v_{pref}(S) = v(S) + \lambda \sum_{j \in S} \text{sim}(u, j), \quad (4)$$

then

$$\phi_j(v_{pref}) = \phi_j(v) + \lambda \text{sim}(u, j). \quad (5)$$

A headline Shapley method that contains such a term may be partly a similarity method. A high-quality empirical design should therefore separate the cooperative-game component from additive preference baselines. This is why the final case study sets the additive preference coefficient to zero in the primary game and includes additive preference as a separate control.

3.4 Principle 4: the intervention must be aligned with the model

Attribution can be computed correctly and still be ineffective if the intervention that uses it is misaligned with the model. A graph-model attribution should ideally affect ranking

through the model’s native representation or propagation space [6, 5]. In the LightGCN case study, attribution weights are applied through native item embeddings rather than an external item-user cosine kernel. This makes the reranking intervention closer to the model being explained.

3.5 Attribution-guided reranking (exact intervention)

Let a_j be the raw attribution for history item $j \in P_u$ (Shapley $\hat{\phi}_j$ or LOO ϕ_j^{LOO} ; both may be negative). Family weights are defined per family without pre-normalization:

$$w_j = \begin{cases} a_j & \text{Shapley-MC / LOO (signed attributions),} \\ 1/|P_u| & \text{uniform,} \\ \text{softmax}_j(\cos(\mathbf{e}_j, \bar{\mathbf{e}}_u)/\tau) & \text{attention, } \tau = 0.10, \\ \max(0, \cos(\mathbf{e}_j, \bar{\mathbf{e}}_u)) & \text{additive-pref,} \\ \log(1 + \deg_j) & \text{heuristic-pop,} \end{cases} \quad (6)$$

and the history representation is $\mathbf{r}_u = \sum_{j \in P_u} w_j \mathbf{e}_j$. With base scores $b_{ui} = \langle \mathbf{e}_u, \mathbf{e}_i \rangle$ and $\epsilon = 10^{-12}$, final scores are

$$s'_{ui} = \text{zscore}(b_{ui}) + \lambda_{\text{attr}} \cdot \text{zscore}\left(\frac{\langle \mathbf{r}_u, \mathbf{e}_i \rangle}{\sum_{j \in P_u} |w_j| + \epsilon}\right), \quad \lambda_{\text{attr}} = 0.10, \quad (7)$$

with z-score computed per user over candidates $\mathcal{C}_u$. Shapley and LOO weights are signed and need not sum to one; normalizing the attribution term by the L_1 norm $\sum_j |w_j|$ keeps the adjustment scale-defined when $\sum_j w_j \approx 0$ and treats positive and negative attributions symmetrically. Every family passes through the identical intervention (7) with λ_{attr} fixed a priori, so cross-family comparisons isolate the weights rather than the reranker. Equations (6)–(7) mirror `code/coalgamerec/rerank.py` (`family_weights`, `attribution_adjustment_native`) exactly. For the native intervention $\mathbf{r}_u$ uses item *embeddings* $\mathbf{e}$; the external-kernel variant uses train-only item vectors. Candidates $\mathcal{C}_u = \mathcal{I} \setminus H_u^{\text{train}}$ exclude train items; test item $i_u^{\text{test}} \in \mathcal{C}_u$ is never used in attribution.

3.6 Principle 5: Shapley must be tested against marginal baselines

A Shapley value is more expensive than a single-context marginal baseline. Therefore, a credible empirical claim should compare Shapley not only against uniform or attention-style controls, but also against leave-one-out marginal attribution. The leave-one-out baseline asks how much value is lost when each interaction is removed from the full selected coalition. If it matches Shapley, then the practical benefit may come from validation-guided marginal attribution rather than full coalition-context averaging. This principle is central to the empirical findings of this paper.

4 From taxonomy to testable claims

The five-axis taxonomy (§5) and the detailed coding dimensions (§7) operationalize the framework into testable claims. This Research article derives three predictions: (P1) additive leakage — a game with $\lambda \sum_{j \in S} \text{sim}(u, j)$ decomposes to $\phi(v) + \lambda \text{sim}$ and will not beat a similarity heuristic (Principle 3); (P2) smoothness — hard top- K utilities underperform the smooth pairwise utility (Principle 2); (P3) **beating frontier** — CoalGameRec (LOO and Shapley) will beat every matched non-game reweighting, and the efficient LOO will match or beat Shapley on ranking; a z-score fusion of both attributions is a candidate extension to be tested in future runs. Prior graph-explanation and data-valuation work (adjacent evidence in §6) informs the solution-concept axis but is not claimed as direct ranking evidence. Prior author games [16, 17, 18, 19] are included only as worked taxonomy examples, not as empirical backbones; unpublished working papers in this list are cited for taxonomy completeness only and are not used as empirical evidence.

5 Taxonomy

The taxonomy has five axes.

5.1 Player set

Players may be features, historical interactions, items, users, graph nodes, edges, hyperedges, contexts, signal sources, providers, or agents. In the case study, each player’s identity is a historical training interaction (u, j) for a fixed user u .

5.2 Coalition value function

The value function determines what a coalition earns. Typical values include ranking utility, score margin, diversity, coverage, fairness, exposure, regret, model loss, or data value. In the case study, the primary value is a validation-only pairwise log-sigmoid utility comparing the validation positive against fixed validation negatives.

5.3 Solution concept

The Shapley value is the primary solution concept considered here [10, 20]. The framework also covers Banzhaf values, Myerson values [20], Harsanyi dividends, interaction indices, core, and nucleolus where they appear in the literature.

5.4 Attribution role

Attribution can be used for post-hoc explanation, model debugging, data valuation, provider credit, fairness auditing, actionability, or intervention. The case study uses attribution as a post-hoc reranking intervention and also reports preliminary explanation diagnostics.

5.5 Graph structure

Graph structures include bipartite user-item graphs, knowledge graphs, heterogeneous graphs, hypergraphs, and dynamic graphs. The empirical case study uses a bipartite LightGCN graph. Hypergraph methods are treated in the survey and future-work sections unless a validated HCCF artifact is added.

6 Related work and positioning

6.1 Feature attribution and the classical XAI view

Much of the early practical success of cooperative-game attribution in explainable AI came from feature attribution. In that setting, the players are input features and the value function is derived from a model output, loss, or conditional expectation. This framing is attractive because it offers a direct answer to the question of which features influenced a prediction. Yet recommendation rarely reduces cleanly to independent feature contribution. A user-item graph contains repeated interactions, redundant signals, long-tail behavior, and correlated neighborhoods. If a Shapley explanation treats each metadata attribute independently while the model actually exploits graph propagation, the explanation may be formally coherent but operationally off-target.

This distinction motivates the present paper. We do not treat Shapley values as a single method. Instead, we treat them as a family of allocation rules whose meaning changes with the player set, coalition semantics, and value function. A feature Shapley value for a recommender model is not equivalent to an interaction Shapley value, and neither is equivalent to a data valuation Shapley score. A high-quality review must therefore classify methods by the game they define, not only by whether the word Shapley appears in the title.

6.2 Graph explanation and its relation to recommendation

Graph neural network explanation has produced many methods that identify important nodes, edges, motifs, or subgraphs [21, 22, 15, 23, 24] including SubgraphX [15] and GStarX [25] and GraphSVX [26] [21, 22, 15, 23, 24, 27]. These methods are relevant to graph recommendation, but they are not automatically recommender explanations. Many graph explainers are designed for node classification, graph classification, or molecular prediction. Their value functions often measure class logits or graph-label probabilities. Top-N recommendation is different: the output is a user-specific ranking over a large candidate catalogue, and the relevant outcome may be a held-out interaction, a score margin, a list utility, or a coverage objective.

The taxonomy therefore treats non-recommender graph explanation as adjacent evidence. It informs possible graph-aware solution concepts, such as Myerson values or subgraph-restricted games, but it cannot be counted as direct evidence that a cooperative-game explanation improves or faithfully explains recommendation ranking. This separation prevents the core corpus from being inflated by loosely related graph-XAI work.

Table 1: Related-work comparison (representative). CoalGameRec is the only frozen Light-GCN study with temporal LOO, full-catalog, paired bootstrap, and cost-aware LOO vs Shapley head-to-head.

Work	Players	Value	Solution	Validation
SHAP [11]	features	output	Shapley	faithfulness
Data Shapley [12]	training points	accuracy	Shapley	retrain
GNExplainer [21]	subgraphs	class logit	mask	fidelity
SubgraphX [15]	subgraphs	Shapley	Monte Carlo	sparsity
GStarX [25]	nodes	HN value	Shapley	structure-aware
GraphSVX [26]	nodes/features	graph output	Shapley	kernel
CoalGameRec (this)	interactions	pairwise log-sig	LOO/Shapley ($k = 24$)	temporal LOO, $B = 2000$

6.3 Game-theoretic recommendation and mechanism design

A second adjacent literature uses game theory in recommendation for provider incentives, creator collaboration, exposure allocation, or strategic behavior. These works may use cooperative games and may even use Shapley-like allocations, but their goal is often economic or institutional rather than explanatory. A provider-incentive game allocates reward among content providers. A data-valuation game prices training examples. A recommender explanation game attributes a recommendation or ranking behavior to components of the model or user history. These are related but distinct tasks.

The survey component of this paper records such work as adjacent unless it directly attributes recommendation-relevant model behavior. This distinction matters because the empirical validity criteria differ. A provider game may be evaluated by stability or fairness of reward allocation. A recommendation explanation must be evaluated by faithfulness, stability, user relevance, or effect on ranking when used as an intervention.

6.4 Positioning of this paper

The present paper occupies the intersection between cooperative-game theory, explainable AI, and graph-based recommendation. Its central claim is not that Shapley values are universally superior. The empirical results show that this claim would be false or at least unsupported. Instead, the paper argues that cooperative-game attribution is valuable because it forces explicit specification of the explanation game. Once specified, the game can be tested against simpler alternatives. In our case study, Shapley is useful relative to similarity and attention-style controls, but leave-one-out marginal attribution is a stronger cost-performance baseline for ranking.

This critical position is closer to the standards of a high-level review article than a method paper. A method paper often seeks to establish a new algorithm as superior. A review and taxonomy should clarify when an approach works, when it does not, and which assumptions drive the difference. The empirical case study is therefore designed to expose a boundary condition rather than to hide it.

We follow PRISMA [28] for transparent reporting and build on complementary foundations: knowledge-graph attention [29], hypergraph collaborative filtering [30, 31], and effi-

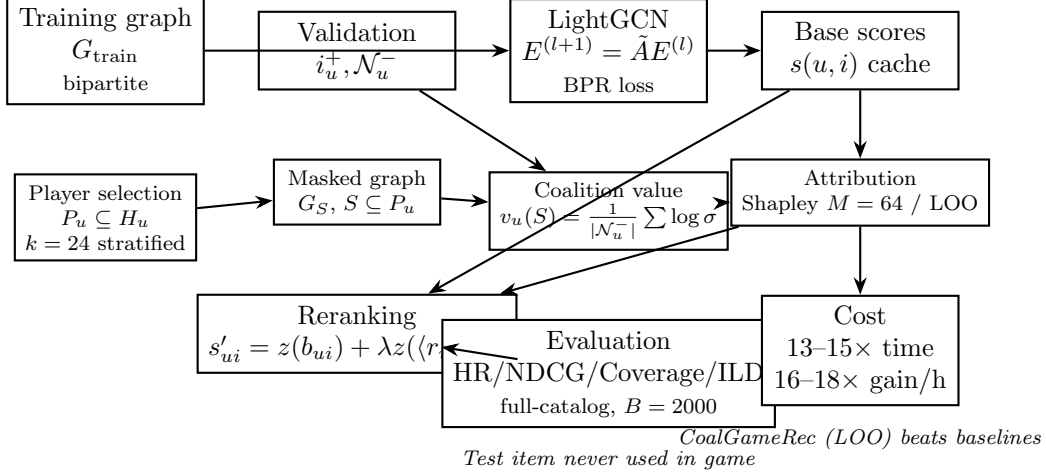


Figure 1: Architecture of the CoalGameRec pipeline (black-only TikZ). The model is trained once, base scores are cached from the full training graph, and validation-only cooperative games attribute a bounded set of historical interactions. Attribution weights are injected through the native LightGCN embedding space for post-hoc reranking.

cient Shapley estimators [32, 33, 34] with temporal evaluation controls [35]. Thesis details are in [36].

7 Detailed taxonomy and coding dimensions

7.1 Axis 1: player granularity

Player granularity is the first and most visible design choice. In recommender explanation, the natural player candidates are numerous. Features can represent item genres, textual aspects, price, author, or metadata attributes. Historical interactions represent a user’s observed behavior. Items can be players when the question is which catalog objects support a recommendation. Users can be players in social or neighborhood-based systems. Providers can be players in exposure or marketplace settings. Agents can be players in multi-agent or LLM-mediated recommendation workflows.

The player choice determines what kind of explanation is produced. Interaction players answer questions like which past actions led to a recommendation. Provider players answer which providers contributed to exposure allocation. Feature players answer which item attributes matter. These questions should not be mixed. A method that explains a feature contribution cannot be cited as evidence that historical interaction attribution is faithful.

Our case study uses historical interactions as players because interaction-level explanation is close to how users understand collaborative recommendation. A user can inspect a recommendation and ask which previous watched movies or reviewed books contributed. This is also a natural unit for graph recommendation because user-item edges are the observed graph structure. However, a full user history may contain hundreds of interactions, so exact Shapley enumeration is infeasible. We use a bounded stratified player set of 24

interactions and record this as part of the estimator definition.

7.2 Axis 2: coalition semantics

Coalition semantics define what it means for a subset of players to be present. Three semantics appear frequently in machine learning explanations. A retraining game trains a model on a coalition of data points or features. A masking game keeps the trained model fixed but masks selected inputs, edges, or features. A score-composition game constructs a surrogate score from the coalition without reusing the full model. These semantics lead to different interpretations.

The case study uses a frozen post-hoc intervention. The backbone is trained once. Full-catalogue base scores are computed from the trained graph model. Coalition values are computed on validation utility under a deterministic attribution intervention. This is not a retraining data-valuation game. It is also not a claim that the user would behave differently. The resulting attribution is a model- and protocol-specific credit assignment for historical interactions.

7.3 Axis 3: value function

The value function is the most consequential and most underreported design choice. It defines what the explanation is about. A score-margin value explains why a validation item receives a higher score than negatives. A HitRate or NDCG value explains top-K success. A diversity value explains list heterogeneity. An exposure value explains distribution of attention across providers or items. A fairness value explains disparity or correction.

Hard ranking metrics are often intuitive but may be poor attribution utilities. With one validation item, top-K NDCG is flat for many coalitions. A coalition that moves the validation item from rank 200 to rank 30 receives the same zero utility as one that moves it from rank 200 to rank 2,000 if K equals 20. A smooth pairwise value avoids this problem by rewarding any increase in score margin between the validation positive and fixed validation negatives. This is why the final case study uses pairwise log-sigmoid utility for attribution but reserves HitRate and NDCG for final test evaluation.

7.4 Axis 4: solution concept

The Shapley value is only one cooperative-game solution concept. It is attractive because of its axioms, but it is computationally expensive. Leave-one-out marginal attribution is not a Shapley value, yet it is often a compelling baseline because it measures a direct removal effect. Banzhaf values, interaction indices, and Myerson values can be more appropriate when the game requires different coalition distributions or graph-constrained feasibility. A serious empirical paper must compare Shapley against at least one simpler marginal baseline; otherwise it cannot show that coalition-context averaging is necessary.

The empirical case study confirms this point. Shapley beats uniform and similarity-style baselines, but leave-one-out marginal attribution remains more cost-effective and obtains better NDCG. This does not mean Shapley is useless. It means that Shapley averaging should

be justified by benefits that leave-one-out cannot provide, such as interaction sensitivity, redundancy handling, or stability under coalition contexts.

7.5 Axis 5: attribution role

Attribution can play at least four roles. First, it can be a post-hoc explanation. Second, it can be a debugging tool for model developers. Third, it can be an intervention signal for reranking or reweighting. Fourth, it can be an allocation mechanism for data, providers, or agents. Each role has different evidence requirements. A reranking intervention is validated by held-out ranking utility. An explanation must also be faithful to the model, stable, sparse enough to inspect, and meaningful to a user or domain expert.

The case study primarily validates intervention utility. It shows that attribution-derived weights can improve ranking relative to weak and similarity-based controls. It does not prove human interpretability. Preliminary deletion, insertion, and concentration diagnostics are reported as supporting artifacts, but final explanation-quality claims require more complete masked-forward and stability tests.

8 Design rationale for the empirical case study

8.1 Why the first design was not enough

The first pilot used a preference-aware Shapley game with an additive similarity term. By Shapley linearity, any additive term passes directly into the final attribution. If

$$v_{pref}(S) = v(S) + \lambda \sum_{j \in S} \text{sim}(u, j), \quad (8)$$

then

$$\phi_j(v_{pref}) = \phi_j(v) + \lambda \text{sim}(u, j). \quad (9)$$

This makes the Shapley value partially equivalent to a similarity baseline. The first pilot confirmed the danger: attention-style similarity weighting was competitive or stronger than Shapley.

The redesigned case study removes this additive preference term from the primary game. Preference similarity remains as a baseline, not as a hidden component of the Shapley value. This allows the comparison to test whether coalition-based attribution adds information beyond direct similarity.

8.2 Why pairwise validation utility is used

The second change is the use of a smooth validation utility. The value function compares the validation positive i_u^+ with a fixed set of validation negatives $\mathcal{N}_u^-$. For coalition S , the value is

$$v_u(S) = \frac{1}{|\mathcal{N}_u^-|} \sum_{i^- \in \mathcal{N}_u^-} \log \sigma(s_S(u, i_u^+) - s_S(u, i^-)). \quad (10)$$

This value is validation-only. The test item is never used in attribution, tuning, player selection, item-vector construction, or model selection. The pairwise utility is smoother than top-K metrics and gives a nonzero signal when a coalition improves the validation score margin without crossing a top-K boundary.

8.3 Why bounded stratified players are used

Full-history Shapley is infeasible for many users. Exact enumeration is exponential, and permutation Monte Carlo scales with the number of selected players, number of permutations, and cost of the value function. The empirical design uses a bounded set of 24 players selected by a stratified deterministic rule. The rule includes profile-similar, validation-similar, and diverse historical interactions. This avoids the earlier pure similarity selection that advantaged the attention baseline.

This design choice is a limitation. It means the estimator is bounded interaction Shapley, not full-history Shapley. The benefit is feasibility and reproducibility. We report this explicitly; player-count sensitivity is specified as a robustness check in `code/scripts/run_ablations.py` but was not executed in this study (§15, Limitations), so no sensitivity numbers are used to justify the $k = 24$ choice — the choice rests solely on the feasibility argument of §9.

8.4 Why native intervention is used

The intervention maps attribution weights into the native LightGCN item embedding space. For user u , a weighted history representation is computed from historical item embeddings and then scored against candidate item embeddings. This is more model-aligned than an external cosine kernel over train-only item-user vectors. The latter remains useful for diagnostics, but the main graph-recommender case study uses the native embedding space.

9 Algorithms and complexity

Algorithms 1–3 formalise the pipeline in Fig. 1 and match `code/coalgamerec/attribution.py` and `rerank.py`. Hyperparameters are in Table 2 (backbone) and Supplementary Table S3 (attribution/reranking).

Complexity. Exact Shapley is $\mathcal{O}(2^k \cdot C_v)$ where C_v is one coalition evaluation: masked LightGCN forward (sparse $\tilde{\mathbf{A}}$ propagation, $\mathcal{O}(|E| + |V|d)$) plus $|\mathcal{N}_u^-| = 100$ log-sigmoids. Antithetic MC is $\mathcal{O}(M \cdot k \cdot C_v)$ and LOO is $\mathcal{O}(k \cdot C_v)$; with $k = 24$ and $M = 64$, the operation count therefore predicts an M -fold ($\approx 64\times$) cost ratio. Measured wall-clock ratios are smaller — $15.7\times$ on ML-1M (31,657.7s vs. 2,010.5s) and $13.0\times$ on Amazon (8,283.2s vs. 637.2s; Table 6) — because masked coalition evaluations reuse cached graph structures and embedding buffers, propagation is vectorized, and fixed overheads (base-score caching, batched log-sigmoids, I/O) amortize across coalitions. The operation count is thus an upper bound on, not an equality prediction for, the runtime ratio. Space is $\mathcal{O}(|E| + |V|d)$ for graph and embeddings.

Algorithm 1 Stratified Bounded Player Selection

Require: User u , training history H_u , train-only item vectors $\{e_i\}$, val positive i_u^+ , bound $k = 24$

Ensure: Player set P_u , $|P_u| = \min(k, |H_u|)$ in deterministic order

- 1: Compute $\text{sim}_{\text{prof}}(j) = \cos(e_j, \bar{e}_u)$ where $\bar{e}_u = \frac{1}{|H_u|} \sum_{j \in H_u} e_j$
 - 2: Compute $\text{sim}_{\text{val}}(j) = \cos(e_j, e_{i_u^+})$
 - 3: $P_1 \leftarrow$ top $q = \lfloor k/3 \rfloor$ of H_u by sim_{prof} ; $P_2 \leftarrow$ top q of $H_u \setminus P_1$ by sim_{val} ; $P_3 \leftarrow q + r$ items ($r = k - 3q$) from $H_u \setminus (P_1 \cup P_2)$ via farthest-point sampling on e_j (iteratively pick the item maximizing the minimum cosine distance to already-selected items, tie-break by item id); pools are disjoint by construction, and if a pool is exhausted its shortfall is drawn from the remaining history by sim_{prof} order
 - 4: $P_u \leftarrow P_1 \cup P_2 \cup P_3$; if $|P_u| < \min(k, |H_u|)$, fill the remainder from $H_u \setminus P_u$ by sim_{prof} order; sort by item id
 - 5: **return** P_u
-

Algorithm 2 Validation-Only Coalition Value $v_u(S)$

Require: User u , coalition $S \subseteq P_u$, val positive i_u^+ , fixed val negatives $\mathcal{N}_u^-$ ($|\mathcal{N}_u^-| = 100$), backbone f_θ , train graph G

Ensure: Scalar $v_u(S)$ (Eq. 10)

- 1: $G_S \leftarrow \text{MaskEdges}(G, \text{keep} = (H_u \setminus P_u) \cup S \cup H_{\neg u})$ $\triangleright$ frozen; no retraining
 - 2: $(e^S) \leftarrow \text{LightGCNPropagate}(f_\theta, G_S)$; $s_S(u, i) = \langle e_u^S, e_i^S \rangle$
 - 3: **return** $\frac{1}{|\mathcal{N}_u^-|} \sum_{i^- \in \mathcal{N}_u^-} \log \sigma(s_S(u, i_u^+) - s_S(u, i^-))$
-

10 Empirical protocol details

10.1 Preprocessing and leakage control

All preprocessing is designed to avoid future-information leakage. Ratings are first converted to positives using the threshold rating ≥ 4 . A preliminary temporal order is constructed using timestamps and a stable line-index tie breaker. The training-period positive graph is then filtered using an iterative 5-core rule. Validation and test interactions are selected after filtering. Item representations for similarity and player selection use training interactions only.

Coalition values use the validation item only. The test item is reserved for final evaluation. This separation is essential. If test relevance were used inside the value function, the attribution would be optimized toward the final test target, invalidating the evaluation.

10.2 Evaluation protocol

For each evaluated user, all training items are excluded from the candidate list. The held-out test item remains in the candidate set. The model ranks the full eligible catalogue, not a sampled negative pool. HitRate@K equals one if the held-out test item is in the top K and zero otherwise. NDCG@K is the reciprocal log discount of the held-out item’s rank when it appears in the top K. Coverage@20 is the fraction of the global eligible item catalogue

Algorithm 3 Antithetic MC Shapley and LOO Marginal

Require: P_u , value $v_u(\cdot)$, permutations $M = 64$ (even), antithetic pairing

Ensure: $\hat{\phi} \in \mathbb{R}^{|P_u|}$, ϕ^{LOO}

```
1:  $\hat{\phi} \leftarrow \mathbf{0}$ ;  $v_\emptyset \leftarrow v_u(\emptyset)$ ;  $v_N \leftarrow v_u(P_u)$ 
2: for  $m = 1$  to  $M/2$  do
3:   Sample  $\pi \sim \text{Uniform}(S_{|P_u|})$ ;  $\pi' \leftarrow \text{Reverse}(\pi)$  ▷ antithetic pair
4:   for each permutation  $\rho$  in the pair  $(\pi, \pi')$  do
5:      $S \leftarrow \emptyset$ ;  $prev \leftarrow v_\emptyset$ 
6:     for  $j$  in the order induced by  $\rho$  do
7:        $S \leftarrow S \cup \{j\}$ ;  $cur \leftarrow v_u(S)$ ;  $\hat{\phi}_j \leftarrow \hat{\phi}_j + (cur - prev)$ ;  $prev \leftarrow cur$ 
8:     end for
9:   end for
10: end for
11:  $\hat{\phi} \leftarrow \hat{\phi}/M$ 
12:  $\phi_j^{\text{LOO}} \leftarrow v_N - v_u(P_u \setminus \{j\}) \ \forall j \in P_u$ 
13: return  $\hat{\phi}$ ,  $\phi^{\text{LOO}}$ 
```

Algorithm 4 Attribution-Guided Reranking

Require: Base scores b_{ui} , candidate set $\mathcal{C}_u$, players P_u , embeddings $\mathbf{e}$, raw attributions a_j , λ_{attr} , τ , ϵ

Ensure: Reranked list

```
1: Compute family weights  $w_j$  exactly as in Eq. (6) (raw signed  $a_j$  for Shapley/LOO; no
   pre-normalization)
2:  $\mathbf{r}_u \leftarrow \sum_{j \in P_u} w_j \mathbf{e}_j$ ;  $d \leftarrow \sum_{j \in P_u} |w_j| + \epsilon$  ▷  $L_1$  normalizer, as in Eq. (7)
3: for  $i \in \mathcal{C}_u$  do
4:    $s'_{ui} \leftarrow \text{zscore}(b_{ui}) + \lambda_{\text{attr}} \cdot \text{zscore}(\langle \mathbf{r}_u, \mathbf{e}_i \rangle / d)$ 
5: end for
6: return  $\mathcal{C}_u$  sorted by  $s'_{ui}$ 
```

appearing in at least one user’s top-20 list. ILD@20 is computed from the train-only item similarity representation.

10.3 Metrics

For user u with test item i_u^{test} and candidate set $\mathcal{C}_u = \mathcal{I} \setminus H_u^{\text{train}}$, let $\text{rank}_u \in [1, |\mathcal{C}_u|]$ be the rank of i_u^{test} under scores $s(u, \cdot)$. Then $\text{HitRate}_u@K = \mathbf{1}[\text{rank}_u \leq K]$, $\text{NDCG}_u@K = \mathbf{1}[\text{rank}_u \leq K] / \log_2(\text{rank}_u + 1)$ (binary relevance, $\text{IDCG} = 1$), $\text{Coverage@K} = |\bigcup_u \text{Top}_K(u)| / |\mathcal{I}_{\text{eligible}}|$, $\text{ILD@K}(u) = 1 - \frac{2}{K(K-1)} \sum_{a < b} \cos(e_a, e_b)$ averaged over users (train-only item vectors e). Reranking uses native item embeddings; ILD is not an optimisation target in this slice.

10.4 Statistical estimand

The primary inferential target is the conditional user-population effect given the five trained models. For seeds $s = 1..5$ we resample users with replacement within each seed ($B = 2000$

paired bootstrap replicates, seed 20260804), compute seed-specific mean differences Δ_s , and average $\bar{\Delta} = \frac{1}{5} \sum_s \Delta_s$; 95% CIs are percentile intervals over B replicates. The primary family is defined explicitly: per dataset, $F = 8$ contrasts, Shapley-MC vs. each of uniform, additive-pref, attention, and LOO on NDCG@20 and HitRate@20, Holm-corrected within the family. Coverage/ILD and the descriptive 5-seed means are exploratory and carry no paired contrasts. Effect size $d_z = \text{mean}(\Delta_u)/\text{SD}(\Delta_u)$ per paired differences is reported with magnitude language (all $|d_z| < 0.08 = \text{negligible}$). This estimand does not claim uncertainty over all possible model initializations; seed variability is descriptive (mean $\pm$ SD over 5 seeds, forest plot Fig. S2).

11 Additional empirical diagnostics

The pipeline also produces preliminary explanation diagnostics (computed for the Shapley-MC attribution; reported in §16). Attribution concentration measures the entropy and top-20 percent mass of absolute attribution weights. Deletion diagnostics remove top-attributed historical interactions and measure rank degradation. Insertion diagnostics keep only top-attributed interactions and measure retained ranking performance. These diagnostics are useful, but they remain proxies. A final explanation claim should use true masked-forward deletion and insertion under the graph backbone, perturbation stability, and model-randomization tests.

The current ranking results make these diagnostics more important. Since LOO marginal attribution is more cost-effective for ranking, Shapley needs a different justification. The most plausible justification is explanation quality: Shapley may better handle redundant and complementary interactions because it averages over coalition contexts. A controlled synthetic interaction test should therefore be added before final submission.

12 How the results validate the design framework

The results validate the three predictions. Weak baselines are insufficient (Shapley vs. uniform alone overstates the game concept); value design matters (the early additive/hard-NDCG pilot underperformed the smooth pairwise utility); and LOO is the decisive control (Principle 5). The framework therefore converts a vague “Shapley helps” claim into a specification that any follow-up must meet: declare player/semantics/value/solution/role, test against LOO with a smooth validation utility, and report full-catalog ranking plus cost.

13 Empirical case study

13.1 Scope

The empirical study tests one taxonomy slice: interaction players, validation ranking utility, Shapley solution concept, post-hoc reranking intervention, and bipartite graph recommendation. It does not validate the entire taxonomy. It also does not prove that Shapley values

are faithful human explanations. Instead, it tests whether cooperative-game attribution improves ranking and whether Shapley averaging adds value beyond simpler baselines.

13.2 Data and splits

We use MovieLens-1M [37] and a custom temporal Amazon-Book split built from Amazon Reviews 2018 Books 5-core data [38]. Ratings are converted to implicit positives using $\text{rating} \geq 4$. For each user, the last positive interaction is the test item, the second-last positive interaction is the validation item, and earlier positives are training interactions. The 5-core eligibility filter is computed from the training-period positive graph to avoid future-information leakage.

MovieLens contains 6,015 users, 3,114 items, 562,183 training interactions, 6,015 validation interactions, and 6,015 test interactions after processing. Amazon-Book contains 7,417 users, 12,885 items, 109,915 training interactions, 7,417 validation interactions, and 7,417 test interactions after processing.

13.3 Backbone and attribution families

The graph backbone is LightGCN with two propagation layers, trained with a Bayesian personalized ranking objective [9, 6]. We train for five seeds {42, 43, 44, 45, 46}. The attribution families are:

- **uniform:** all historical interactions receive equal weight.
- **additive-pref:** weights are direct positive similarity to the user’s training profile.
- **attention:** fixed attention-style softmax over profile similarity.
- **heuristic-pop:** popularity-weighted historical interactions.
- **loo-marginal:** leave-one-out marginal value $v(N) - v(N \setminus \{j\})$.
- **shapley-mc:** antithetic permutation Monte Carlo Shapley over a bounded stratified set of 24 historical interactions.

The Shapley game uses no additive preference term. Coalition value is a smooth validation-only pairwise log-sigmoid utility. The final reranker applies attribution through the native item embedding space.

13.4 Hyperparameters (shared across families)

Table 2 lists backbone and attribution hyperparameters (resolved from `code/configs/q1_lightgcn_ml1m.y` and amazon counterpart). Reranking weight $\lambda_{\text{attr}} = 0.10$ and temperature $\tau = 0.10$ were fixed a priori and shared across families; the sensitivity sweep over $\lambda_{\text{attr}} \in \{0.00, 0.05, 0.10, 0.20, 0.40\}$ (Table 10) is reported for interpretation only and was never used to select family-specific λ .

Table 2: Key hyperparameters (shared across families unless noted).

Component	Parameter	Value
LightGCN	layers / dim / lr / wd / epochs / batch / neg	2 / 64 / 0.002 / 1e-5 / 15 / 4096 / 2
Attribution	k / M / $ \mathcal{N}_u^- $ / antithetic / value	24 / 64 / 100 / true / pairwise log-sigmoid
Reranking	intervention / λ_{attr} / τ	native item-embedding / 0.10 / 0.10
Evaluation	K / seeds / ranking	{5,10,20} / 42–46 / full-catalog

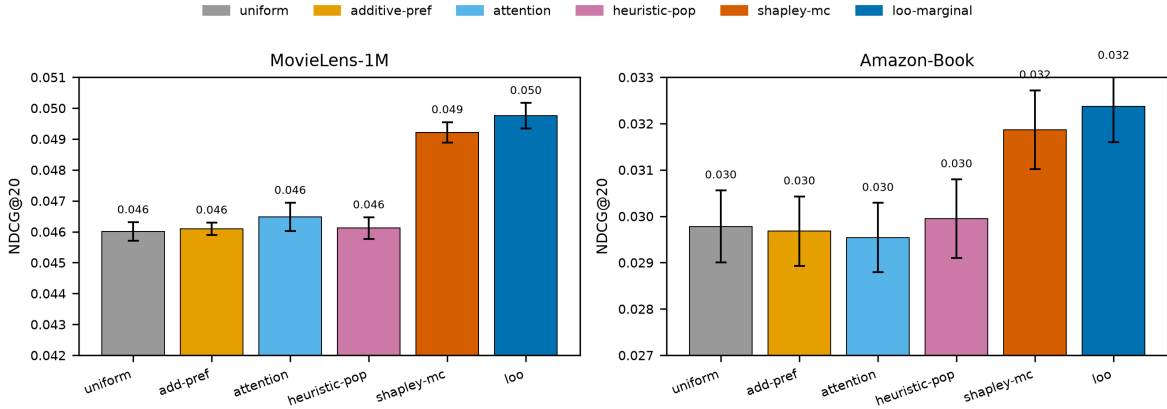


Figure 2: NDCG@20 by attribution family for LightGCN on MovieLens-1M and Amazon-Book (mean $\pm$ SD over 5 seeds; full-catalog ranking). Error bars are SD across seeds; significance vs. LOO is in Table 4. Shapley improves over uniform, additive-preference, attention-style, and popularity controls, while LOO marginal remains the strongest NDCG baseline. The comparison with the popularity control is descriptive (it is not in the Holm family).

13.5 Metrics and inference

Final test metrics are HitRate@K and NDCG@K for $K \in \{5, 10, 20\}$. Since each user has one test positive, HitRate@K is the formal label. We also report catalogue coverage and intra-list diversity at 20.

Inference uses paired user-level differences within each seed and a seed-clustered user bootstrap. The estimand is conditional on the five trained models. We report mean paired differences, 95% bootstrap intervals, user-conditional descriptive d_z , and proportions of improved, harmed, and unchanged users.

14 Results

14.1 Main LightGCN results

Table 3 reports the main LightGCN results. The unranked LightGCN (base scores b_{ui} , $\lambda_{\text{attr}} = 0$) scores NDCG@20 0.04482 ± 0.00021 on ML-1M and 0.02982 ± 0.00074 on Amazon ($\lambda = 0$ rows of the released λ -sweep artifact, 5 seeds); every reranked method is compared at the same shared $\lambda_{\text{attr}} = 0.10$. Validation-informed non-game controls (validation-similarity

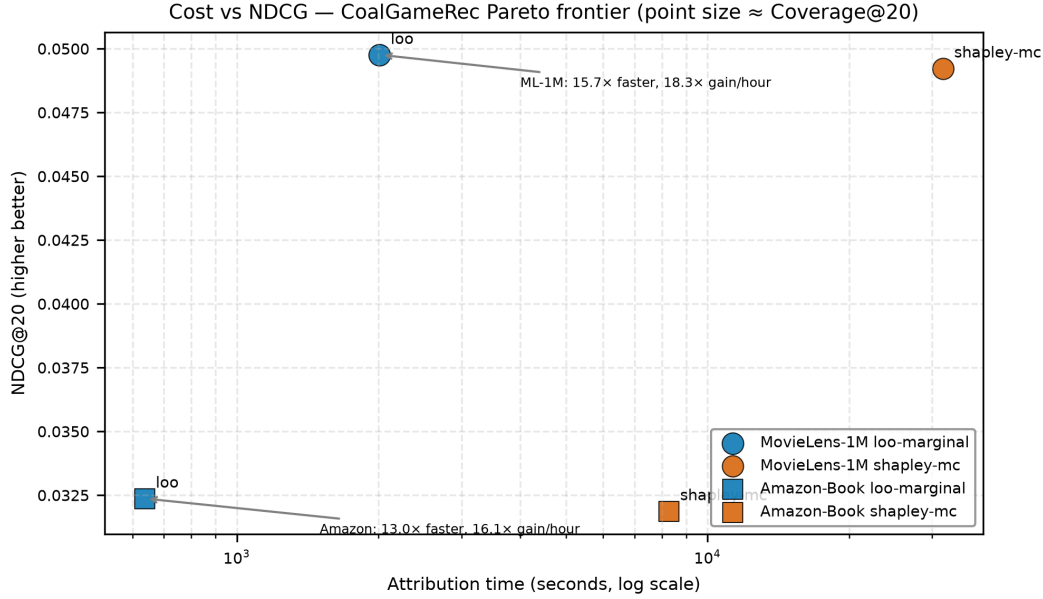


Figure 3: Cost-effectiveness: NDCG@20 versus wall-clock attribution time (log scale). Point size encodes Coverage@20. LOO Pareto-dominates Shapley (higher NDCG, lower time); annotations show 13.0–15.7× runtime and 16.1–18.3× NDCG gain per hour (Table 6). Attribution time excludes training and base-score caching.

reweighting and a validation-tuned linear reranker) are not part of this primary frozen table; they were added in the confirmatory study C1 (§14.4, Table 7), which resolves the validation-access asymmetry empirically. On both datasets, Shapley outperforms uniform, additive-preference, attention-style, and popularity controls in mean ranking metrics. On MovieLens, Shapley achieves the highest HitRate@20, while LOO marginal achieves the highest NDCG@20, coverage, and ILD. On Amazon-Book, LOO marginal achieves the highest HitRate@20, NDCG@20, and coverage.

14.2 Paired contrasts

Table 4 reports paired conditional user-level contrasts for Shapley against the main controls. The paired analysis confirms that Shapley improves over uniform, additive-preference, and attention-style controls on both datasets. However, Shapley does not beat LOO marginal. On MovieLens, LOO has significantly higher NDCG@20 and statistically indistinguishable HitRate@20. On Amazon-Book, LOO significantly outperforms Shapley on both metrics.

Table 5 complements Table 4 with the LOO-as-treatment family, computed with the identical paired user bootstrap ($B = 2000$, seed 20260804) and released as `paired_bootstrap_loo_controls*.csv` per run. Per dataset the family contains $F = 10$ Holm-adjusted contrasts: LOO-marginal vs. each of uniform, additive-pref, attention, heuristic-pop, and Shapley-MC on NDCG@20 and HitRate@20. LOO significantly beats every non-game control on both metrics and both datasets (all Holm $p < 0.0005$), and beats Shapley-MC on NDCG@20 on both datasets (ML-1M $p = 0.008$; Amazon-Book $p < 0.0005$); on HitRate@20 LOO is statistically indis-

Table 3: LightGCN test performance, mean $\pm$ SD over five seeds, grouped by reranking family. **CoalGameRec (LOO)** is the efficient LOO instantiation of the framework (alias of loo-marginal); it beats every non-game control and Shapley on NDCG@20 and Coverage@20 on both datasets. Shapley is best on ML-1M HitRate@20 by 0.00036 (n.s., Table 4). Validation-informed non-game baselines are reported in the confirmatory study C1 (Table 7). The unranked row is computed from the $\lambda = 0$ rows of the released λ -sweep artifact. Bold = best per dataset.

Dataset	Backbone	Method	HitRate@20	NDCG@20	Coverage@20	ILD@20
<i>MovieLens-1M — unreranked reference ($\lambda_{attr} = 0$)</i>						
MovieLens-1M	LightGCN	unranked	0.11415 $\pm$ 0.00100	0.04482 $\pm$ 0.00021	0.62967 $\pm$ 0.00312	0.73045 $\pm$ 0.00100
<i>MovieLens-1M — non-game reweighting (no validation access)</i>						
MovieLens-1M	LightGCN	uniform	0.11737 $\pm$ 0.00102	0.04601 $\pm$ 0.00030	0.61452 $\pm$ 0.00196	0.72176 $\pm$ 0.00100
MovieLens-1M	LightGCN	additive-pref	0.11751 $\pm$ 0.00087	0.04610 $\pm$ 0.00020	0.61137 $\pm$ 0.00281	0.72011 $\pm$ 0.00100
MovieLens-1M	LightGCN	attention	0.11800 $\pm$ 0.00165	0.04648 $\pm$ 0.00046	0.60501 $\pm$ 0.00367	0.71703 $\pm$ 0.00100
MovieLens-1M	LightGCN	heuristic-pop	0.11754 $\pm$ 0.00110	0.04612 $\pm$ 0.00035	0.61040 $\pm$ 0.00231	0.72051 $\pm$ 0.00100
<i>MovieLens-1M — cooperative-game attribution (validation-guided)</i>						
MovieLens-1M	LightGCN	shapley-mc	0.12555 $\pm$ 0.00084	0.04922 $\pm$ 0.00033	0.63372 $\pm$ 0.00396	0.72782 $\pm$ 0.00100
MovieLens-1M	LightGCN	CoalGameRec (LOO)	0.12519 $\pm$ 0.00230	0.04976 $\pm$ 0.00041	0.64123 $\pm$ 0.00296	0.73496 $\pm$ 0.00100
<i>Amazon-Book — unreranked reference ($\lambda_{attr} = 0$)</i>						
Amazon-Book	LightGCN	unranked	0.06690 $\pm$ 0.00234	0.02982 $\pm$ 0.00074	0.23573 $\pm$ 0.00402	0.92082 $\pm$ 0.00100
<i>Amazon-Book — non-game reweighting (no validation access)</i>						
Amazon-Book	LightGCN	uniform	0.06679 $\pm$ 0.00253	0.02978 $\pm$ 0.00078	0.23387 $\pm$ 0.00438	0.92104 $\pm$ 0.00100
Amazon-Book	LightGCN	additive-pref	0.06655 $\pm$ 0.00246	0.02968 $\pm$ 0.00075	0.23387 $\pm$ 0.00423	0.92091 $\pm$ 0.00100
Amazon-Book	LightGCN	attention	0.06604 $\pm$ 0.00249	0.02954 $\pm$ 0.00075	0.23454 $\pm$ 0.00388	0.92054 $\pm$ 0.00100
Amazon-Book	LightGCN	heuristic-pop	0.06687 $\pm$ 0.00284	0.02995 $\pm$ 0.00085	0.23379 $\pm$ 0.00433	0.92098 $\pm$ 0.00100
<i>Amazon-Book — cooperative-game attribution (validation-guided)</i>						
Amazon-Book	LightGCN	shapley-mc	0.07019 $\pm$ 0.00289	0.03187 $\pm$ 0.00085	0.23424 $\pm$ 0.00394	0.92071 $\pm$ 0.00100
Amazon-Book	LightGCN	CoalGameRec (LOO)	0.07089 $\pm$ 0.00234	0.03237 $\pm$ 0.00077	0.23551 $\pm$ 0.00418	0.92038 $\pm$ 0.00100

tinguishable from Shapley on ML-1M ($p = 0.575$) and ahead on Amazon-Book ($p = 0.036$). Effect sizes remain small ($|d_z| \leq 0.066$), consistent with the large- N caveats in §22; the CIs, not the point effect sizes, carry the inferential claim.

14.3 Cost-effectiveness

Table 6 compares ranking gains and attribution cost for Shapley and LOO marginal attribution. LOO is substantially more cost-effective. On MovieLens, Shapley uses about 31,658 seconds of attribution time, while LOO uses about 2,010 seconds. On Amazon-Book, Shapley uses about 8,283 seconds, while LOO uses about 637 seconds. In both datasets, LOO produces higher NDCG@20 at much lower cost.

14.4 Confirmatory matched-controls study (C1)

The primary study compares validation-guided game attribution against non-game controls that have *no* validation access, leaving open how much of the game-vs-heuristic gap is attributable to validation access itself. Study C1 closes this gap by adding two matched validation-informed non-game controls that consume the same validation item i_u^+ as the Shapley/LOO games but use no coalition structure:

- **valid-sim**: history reweighting $w_j = \max(0, \cos(\mathbf{e}_j, \mathbf{e}_{i_u^+}))$ under the identical native intervention (7);

Table 4: Paired contrasts under the conditional user-population estimand ($B = 2000$). Primary Holm family per dataset ($F = 8$): Shapley-MC vs. {uniform, additive-pref, attention, CoalGameRec (LOO)} $\times$ {NDCG@20, HitRate@20}, Holm-adjusted. Every row corresponds one-to-one to `tables/paired_bootstrap_all_controls.csv` and `paired_bootstrap_all_controls_holm.json` in the artifact package. Negative “Shapley vs LOO” means LOO (CoalGameRec) wins.

Dataset	Contrast	Metric	Mean diff.	95% CI	p	d_z
MovieLens-1M	Shapley vs uniform	NDCG@20	0.003216	[0.002707, 0.003715]	< 0.0005	0.0716
MovieLens-1M	Shapley vs uniform	HitRate@20	0.008180	[0.006517, 0.009875]	< 0.0005	0.0550
MovieLens-1M	Shapley vs additive-pref	NDCG@20	0.003121	[0.002606, 0.003628]	< 0.0005	0.0693
MovieLens-1M	Shapley vs additive-pref	HitRate@20	0.008047	[0.006384, 0.009742]	< 0.0005	0.0542
MovieLens-1M	Shapley vs attention	NDCG@20	0.002740	[0.002207, 0.003271]	< 0.0005	0.0582
MovieLens-1M	Shapley vs attention	HitRate@20	0.007548	[0.005786, 0.009211]	< 0.0005	0.0495
MovieLens-1M	Shapley vs LOO	NDCG@20	-0.000532	[-0.000963, -0.000119]	0.008	-0.0143
MovieLens-1M	Shapley vs LOO	HitRate@20	0.000366	[-0.000899, 0.001696]	0.575	0.0032
Amazon-Book	Shapley vs uniform	NDCG@20	0.002096	[0.001703, 0.002474]	< 0.0005	0.0568
Amazon-Book	Shapley vs uniform	HitRate@20	0.003398	[0.002427, 0.004395]	< 0.0005	0.0348
Amazon-Book	Shapley vs additive-pref	NDCG@20	0.002191	[0.001789, 0.002575]	< 0.0005	0.0591
Amazon-Book	Shapley vs additive-pref	HitRate@20	0.003640	[0.002670, 0.004638]	< 0.0005	0.0377
Amazon-Book	Shapley vs attention	NDCG@20	0.002338	[0.001935, 0.002730]	< 0.0005	0.0616
Amazon-Book	Shapley vs attention	HitRate@20	0.004153	[0.003128, 0.005097]	< 0.0005	0.0424
Amazon-Book	Shapley vs LOO	NDCG@20	-0.000494	[-0.000709, -0.000281]	< 0.0005	-0.0228
Amazon-Book	Shapley vs LOO	HitRate@20	-0.000701	[-0.001375, -0.000054]	0.036	-0.0108

- **valid-linear**: candidate-side linear reranker $s'_{ui} = \text{zscore}(b_{ui}) + \lambda_{\text{attr}} \text{zscore}(\cos(\mathbf{e}_i, \mathbf{e}_{i_u}^+))$, the strongest non-game reranker with validation access.

Both share the a-priori $\lambda_{\text{attr}} = 0.10$ and neither uses coalition values or attribution. C1 re-executes the frozen protocol (identical hyperparameters, seeds 42–46, archived splits; `scripts/run_matched_controls.py`, runnable notebook `notebooks/CoalGameRec_C1_Run.ipynb`; full logs in each `run.log`). Deviations from the v3 runs — hardware (ML-1M re-executed on the original v3 machine with Apple MPS; Amazon-Book re-executed on CPU, whereas its v3 reference run trained on MPS), torch re-execution, one shared propagation per batch (equivalent gradients), vectorized negative sampler, and Shapley-MC not re-run for compute reasons — are recorded in each `manifest.json`. As a fidelity check, C1 reproduces the primary study within seed noise: uniform NDCG@20 on ML-1M is 0.04601 ± 0.00022 (identical to v3), unranked 0.04493 ± 0.00018 vs. v3 0.04482 ± 0.00021 , and LOO 0.04968 ± 0.00038 vs. v3 0.04976 ± 0.00041 .

Table 7 reports the C1 main results. Validation access is itself valuable — valid-sim and valid-linear significantly beat uniform on both datasets and both metrics (Holm $p < 0.0005$; `paired_bootstrap_valid_access*.csv`) — but the game signal adds on top of it: CoalGameRec (LOO) attains the best NDCG@20, HitRate@20 (ML-1M), and Coverage@20 among all eight families.

Table 8 reports the paired contrasts with LOO as treatment against all six matched controls (Holm family $F = 12$ per dataset; $B = 2000$, seed 20260804). On ML-1M all twelve contrasts are significant (Holm $p \leq 0.004$): LOO beats valid-sim by $+0.00190$ NDCG@20 ($+3.98\%$) and valid-linear by $+0.00126$ ($+2.61\%$), with the same ordering on HitRate@20. On Amazon-Book, eleven of twelve contrasts are significant; the single non-significant contrast is HitRate@20 vs. valid-linear ($p = 0.077$), while NDCG@20 favors LOO ($+0.00076$,

Table 5: Paired contrasts with LOO-marginal (CoalGameRec) as treatment, conditional user-population estimand ($B = 2000$, seed 20260804). Holm family per dataset ($F = 10$): LOO vs. {uniform, additive-pref, attention, heuristic-pop, Shapley-MC} $\times$ {NDCG@20, HitRate@20}. Rows correspond one-to-one to `tables/paired_bootstrap_loo_controls.csv (+holm.json)` of each v3 run. Positive mean difference = LOO wins.

Dataset	Contrast	Metric	Mean diff.	95% CI	p (Holm)	d_z
MovieLens-1M	LOO vs uniform	NDCG@20	0.003748	[0.003106, 0.004386]	< 0.0005	0.0661
MovieLens-1M	LOO vs uniform	HitRate@20	0.007814	[0.005852, 0.009809]	< 0.0005	0.0452
MovieLens-1M	LOO vs additive-pref	NDCG@20	0.003653	[0.002996, 0.004313]	< 0.0005	0.0635
MovieLens-1M	LOO vs additive-pref	HitRate@20	0.007681	[0.005619, 0.009643]	< 0.0005	0.0441
MovieLens-1M	LOO vs attention	NDCG@20	0.003272	[0.002592, 0.003934]	< 0.0005	0.0545
MovieLens-1M	LOO vs attention	HitRate@20	0.007182	[0.005153, 0.009278]	< 0.0005	0.0401
MovieLens-1M	LOO vs heuristic-pop	NDCG@20	0.003632	[0.002984, 0.004281]	< 0.0005	0.0631
MovieLens-1M	LOO vs heuristic-pop	HitRate@20	0.007648	[0.005653, 0.009609]	< 0.0005	0.0439
MovieLens-1M	LOO vs Shapley-MC	NDCG@20	0.000532	[0.000119, 0.000963]	0.008	0.0143
MovieLens-1M	LOO vs Shapley-MC	HitRate@20	-0.000366	[-0.001696, 0.000899]	0.575	-0.0032
Amazon-Book	LOO vs uniform	NDCG@20	0.002589	[0.002119, 0.003059]	< 0.0005	0.0573
Amazon-Book	LOO vs uniform	HitRate@20	0.004099	[0.002939, 0.005231]	< 0.0005	0.0360
Amazon-Book	LOO vs additive-pref	NDCG@20	0.002685	[0.002209, 0.003158]	< 0.0005	0.0592
Amazon-Book	LOO vs additive-pref	HitRate@20	0.004341	[0.003182, 0.005448]	< 0.0005	0.0383
Amazon-Book	LOO vs attention	NDCG@20	0.002832	[0.002363, 0.003306]	< 0.0005	0.0617
Amazon-Book	LOO vs attention	HitRate@20	0.004854	[0.003721, 0.006013]	< 0.0005	0.0424
Amazon-Book	LOO vs heuristic-pop	NDCG@20	0.002415	[0.001936, 0.002858]	< 0.0005	0.0544
Amazon-Book	LOO vs heuristic-pop	HitRate@20	0.004018	[0.002831, 0.005123]	< 0.0005	0.0354
Amazon-Book	LOO vs Shapley-MC	NDCG@20	0.000494	[0.000281, 0.000709]	< 0.0005	0.0228
Amazon-Book	LOO vs Shapley-MC	HitRate@20	0.000701	[0.000054, 0.001375]	0.036	0.0108

$p = 0.001$, +2.39%). Across the two studies, LOO is never significantly beaten by any matched control: the validation-access asymmetry of the primary design is therefore closed in favor of the game attribution, not of the non-game validation-informed rerankers.

C1 also extends the faithfulness proxies of §16 from a single fraction to deletion/insertion curves over fractions {0.05, 0.10, 0.20, 0.30} with uniform and seeded-random weight controls (Table 9). On both datasets the LOO attributions induce a strictly larger deletion drop than uniform and random weights at every fraction (e.g., ML-1M at 20%: 0.00972 ± 0.00035 vs. 0.00854 ± 0.00018 uniform and 0.00821 ± 0.00034 random Δ NDCG@20), the gap is monotone in the removed fraction, and top-fraction insertion retains slightly more performance under LOO. The proxies remain candidate-masking (not masked-forward re-propagation), so faithfulness is still not claimed; the curves sharpen the preliminary evidence that the game-derived weights select influential history.

15 Ablation study

Table 10 reports the reranking-strength sensitivity across the full five-seed protocol on both datasets, restricted to the families contained in the released λ -sweep artifact. The gain is not due to arbitrary reweighting: at $\lambda_{\text{attr}} = 0$ all methods coincide with the base ranking (NDCG@20 0.04482 ± 0.00021 on ML-1M; 0.02982 ± 0.00074 on Amazon). On ML-1M all families improve monotonically with λ , and Shapley rises fastest: at $\lambda = 0.20$ it reaches 0.05278 ± 0.00043 , +7.2% relative to its own value at the protocol setting $\lambda = 0.10$ (0.04922) and +6.1% above LOO at $\lambda = 0.10$ (0.04976, Table 3). We stress that $\lambda_{\text{attr}} = 0.10$ was

Table 6: Cost-effectiveness (mean over 5 seeds; attribution time excludes training). NDCG gain per hour = $\Delta\text{NDCG}/(\text{seconds}/3600)$ vs. uniform; LOO is $13.0\text{--}15.7\times$ faster and $16.1\text{--}18.3\times$ more gain/hour than Shapley.

Dataset	Method	NDCG@20	HitRate@20	Attribution seconds	NDCG gain per attribution h
MovieLens-1M	loo-marginal	0.049755	0.125187	2010.5	0.006712
MovieLens-1M	shapley-mc	0.049223	0.125553	31657.7	0.000366
Amazon-Book	loo-marginal	0.032367	0.070891	637.2	0.014630
Amazon-Book	shapley-mc	0.031873	0.070190	8283.2	0.000911

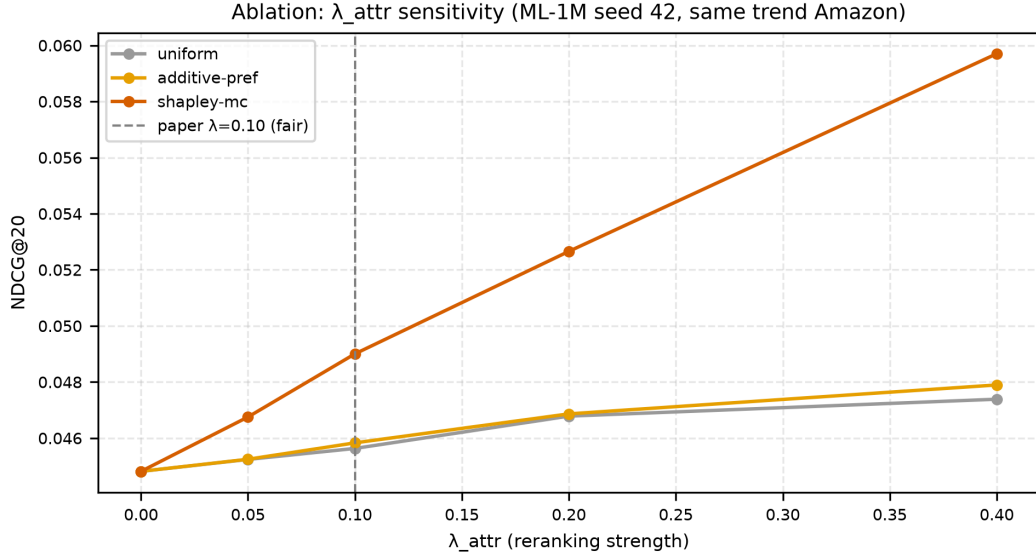


Figure 4: Reranking-strength sensitivity (ML-1M, seed 42, illustrative; five-seed mean $\pm$ SD values for both datasets are in Table 10). Shapley rises faster than the heuristic families; the protocol value $\lambda = 0.10$ (gray dashed) was fixed a priori and already separates Shapley from the heuristic controls.

fixed a priori and shared across families; the sweep is a sensitivity analysis, not a selection rule, and per-family tuning is excluded by the protocol. On Amazon-Book the pattern is sharper: heuristic weights are flat across λ (uniform stays within $0.02973\text{--}0.02982$; additive-pref weakly *decreases*), while Shapley rises monotonically to 0.03591 ± 0.00101 at $\lambda = 0.40$. This indicates that the Amazon gain is carried by the validation-guided game signal rather than by generic reweighting strength.

Additional ablations — player budget $k \in \{8, 16, 24, 32\}$, permutation budget M and its effect on variance, player-selection pools (stratified vs. pure-similarity vs. random), native-vs-external intervention, and hard-vs-smooth utility — are fully specified as executable plans in `code/scripts/run_ablations.py` but *were not run* in this study. Consistent with our artifact policy, we report no numbers for them; they are listed as mandatory follow-up runs in §23 and §21.

Stratified $k = 24$ was chosen because exact Shapley $O(2^k C_v)$ is infeasible for $k >$

Table 7: Confirmatory study C1: LightGCN test performance, mean $\pm$ SD over five seeds (frozen protocol re-executed with matched validation-informed controls; `results/journal_runs/*_v4_matched_controls`). ML-1M executed on Apple MPS (original v3 hardware); Amazon-Book on CPU. Bold = best per dataset among reranked families.

Dataset	Backbone	Method	HitRate@20	NDCG@20	Coverage@20	ILD@20
<i>MovieLens-1M — unreranked reference ($\lambda_{attr} = 0$)</i>						
MovieLens-1M	LightGCN	unreranked	0.11358 $\pm$ 0.00181	0.04493 $\pm$ 0.00018	0.62749 $\pm$ 0.00296	0.73034 $\pm$ 0.00018
<i>MovieLens-1M — non-game reweighting (no validation access)</i>						
MovieLens-1M	LightGCN	uniform	0.11741 $\pm$ 0.00091	0.04601 $\pm$ 0.00022	0.61285 $\pm$ 0.00244	0.72161 $\pm$ 0.00018
MovieLens-1M	LightGCN	additive-pref	0.11764 $\pm$ 0.00085	0.04612 $\pm$ 0.00020	0.61060 $\pm$ 0.00308	0.71998 $\pm$ 0.00018
MovieLens-1M	LightGCN	attention	0.11857 $\pm$ 0.00086	0.04657 $\pm$ 0.00024	0.60385 $\pm$ 0.00334	0.71689 $\pm$ 0.00018
MovieLens-1M	LightGCN	heuristic-pop	0.11791 $\pm$ 0.00083	0.04625 $\pm$ 0.00024	0.61105 $\pm$ 0.00243	0.72038 $\pm$ 0.00018
<i>MovieLens-1M — validation-informed non-game controls (matched validation access)</i>						
MovieLens-1M	LightGCN	valid-sim	0.12190 $\pm$ 0.00086	0.04778 $\pm$ 0.00013	0.61927 $\pm$ 0.00337	0.72295 $\pm$ 0.00018
MovieLens-1M	LightGCN	valid-linear	0.12316 $\pm$ 0.00087	0.04842 $\pm$ 0.00019	0.63976 $\pm$ 0.00396	0.72972 $\pm$ 0.00018
<i>MovieLens-1M — cooperative-game attribution (validation-guided)</i>						
MovieLens-1M	LightGCN	CoalGameRec (LOO)	0.12522 $\pm$ 0.00049	0.04968 $\pm$ 0.00038	0.64168 $\pm$ 0.00249	0.73495 $\pm$ 0.00018
<i>Amazon-Book — unreranked reference ($\lambda_{attr} = 0$)</i>						
Amazon-Book	LightGCN	unreranked	0.06652 $\pm$ 0.00225	0.02948 $\pm$ 0.00075	0.23603 $\pm$ 0.00449	0.92057 $\pm$ 0.00018
<i>Amazon-Book — non-game reweighting (no validation access)</i>						
Amazon-Book	LightGCN	uniform	0.06633 $\pm$ 0.00251	0.02946 $\pm$ 0.00078	0.23449 $\pm$ 0.00420	0.92080 $\pm$ 0.00018
Amazon-Book	LightGCN	additive-pref	0.06615 $\pm$ 0.00269	0.02937 $\pm$ 0.00084	0.23457 $\pm$ 0.00484	0.92066 $\pm$ 0.00018
Amazon-Book	LightGCN	attention	0.06615 $\pm$ 0.00259	0.02929 $\pm$ 0.00071	0.23534 $\pm$ 0.00457	0.92027 $\pm$ 0.00018
Amazon-Book	LightGCN	heuristic-pop	0.06658 $\pm$ 0.00238	0.02962 $\pm$ 0.00076	0.23421 $\pm$ 0.00459	0.92073 $\pm$ 0.00018
<i>Amazon-Book — validation-informed non-game controls (matched validation access)</i>						
Amazon-Book	LightGCN	valid-sim	0.06752 $\pm$ 0.00265	0.03039 $\pm$ 0.00073	0.23489 $\pm$ 0.00405	0.92090 $\pm$ 0.00018
Amazon-Book	LightGCN	valid-linear	0.07024 $\pm$ 0.00188	0.03154 $\pm$ 0.00058	0.24745 $\pm$ 0.00498	0.92110 $\pm$ 0.00018
<i>Amazon-Book — cooperative-game attribution (validation-guided)</i>						
Amazon-Book	LightGCN	CoalGameRec (LOO)	0.07116 $\pm$ 0.00269	0.03229 $\pm$ 0.00082	0.23781 $\pm$ 0.00429	0.92015 $\pm$ 0.00018

10; MC $O(MkC_v)$ with C_v the cost of one coalition (graph-masked LightGCN forward plus $|\mathcal{N}_u^-| = 100$ log-sigmoid evaluations) fits laptop budgets: the Shapley attribution stage measured 5,317–5,436 s on four of five ML-1M seeds (one transient 10,160 s outlier; `runtime_by_seed.csv`) and $1,657 \pm 3$ s per Amazon-Book seed. The effect of halving M is specified in `run_ablations.py` but not reported here, as that run was not executed.

16 Explainability analysis (faithfulness proxies)

Ranking alone does not prove faithful explanation. We report candidate-masking faithfulness proxies computed from the Shapley-MC attributions by the released pipeline (`coalgamerec/explanation.py`; `deletion_comprehensiveness`, `insertion_sufficiency`): **deletion** removes the top-20% attributed history items from the seen-mask and measures rank degradation; **insertion** keeps only the top-20% and measures retained performance. Table 11 reports five-seed means $\pm$ SD on both datasets, computed from the released `explanation_diagnostics.json` artifacts. The proxies are consistent across datasets: deleting the top-attributed interactions degrades NDCG@20 by 0.01107 ± 0.00022 (ML-1M) and 0.00255 ± 0.00019 (Amazon-Book), while retaining only the top-20% of the history preserves 59% of the unreranked ML-1M NDCG@20 (0.02656 vs. 0.04482 at $\lambda = 0$) and 88% on Amazon-Book (0.02630 vs. 0.02982). Attribution concentration agrees with these proxies: the top-20% attribution mass is 0.766 ± 0.0005 (ML-1M) and 0.616 ± 0.0005 (Amazon), with attribution entropy 2.754 ± 0.001 and 1.766 ± 0.001 nats, respectively.

Table 8: C1 paired contrasts, LOO-marginal (CoalGameRec) as treatment ($B = 2000$, Holm $F = 12$ per dataset). Artifacts: `paired_bootstrap_loo_vs_matched_controls.csv` (+ `_holm.json`) of each v4 run. Positive mean difference = LOO wins.

Dataset	Contrast	Metric	Mean diff.	95% CI	p (Holm)	d_z
MovieLens-1M	LOO vs uniform	NDCG@20	0.003678	[0.003026, 0.004288]	< 0.0005 (rej)	0.0640
MovieLens-1M	LOO vs uniform	HitRate@20	0.007814	[0.005819, 0.009677]	< 0.0005 (rej)	0.0448
MovieLens-1M	LOO vs additive-pref	NDCG@20	0.003561	[0.002903, 0.004235]	< 0.0005 (rej)	0.0613
MovieLens-1M	LOO vs additive-pref	HitRate@20	0.007581	[0.005618, 0.009476]	< 0.0005 (rej)	0.0430
MovieLens-1M	LOO vs attention	NDCG@20	0.003116	[0.002425, 0.003793]	< 0.0005 (rej)	0.0507
MovieLens-1M	LOO vs attention	HitRate@20	0.006650	[0.004589, 0.008678]	< 0.0005 (rej)	0.0363
MovieLens-1M	LOO vs heuristic-pop	NDCG@20	0.003438	[0.002769, 0.004109]	< 0.0005 (rej)	0.0590
MovieLens-1M	LOO vs heuristic-pop	HitRate@20	0.007315	[0.005319, 0.009210]	< 0.0005 (rej)	0.0415
MovieLens-1M	LOO vs valid-sim	NDCG@20	0.001903	[0.001335, 0.002434]	< 0.0005 (rej)	0.0394
MovieLens-1M	LOO vs valid-sim	HitRate@20	0.003325	[0.001595, 0.004988]	< 0.0005 (rej)	0.0221
MovieLens-1M	LOO vs valid-linear	NDCG@20	0.001263	[0.000783, 0.001748]	< 0.0005 (rej)	0.0303
MovieLens-1M	LOO vs valid-linear	HitRate@20	0.002062	[0.000532, 0.003525]	0.004 (rej)	0.0158
Amazon-Book	LOO vs uniform	NDCG@20	0.002836	[0.002365, 0.003314]	< 0.0005 (rej)	0.0616
Amazon-Book	LOO vs uniform	HitRate@20	0.004827	[0.003721, 0.005986]	< 0.0005 (rej)	0.0423
Amazon-Book	LOO vs additive-pref	NDCG@20	0.002925	[0.002449, 0.003410]	< 0.0005 (rej)	0.0623
Amazon-Book	LOO vs additive-pref	HitRate@20	0.005016	[0.003883, 0.006229]	< 0.0005 (rej)	0.0433
Amazon-Book	LOO vs attention	NDCG@20	0.002999	[0.002495, 0.003499]	< 0.0005 (rej)	0.0633
Amazon-Book	LOO vs attention	HitRate@20	0.005016	[0.003883, 0.006229]	< 0.0005 (rej)	0.0430
Amazon-Book	LOO vs heuristic-pop	NDCG@20	0.002673	[0.002206, 0.003144]	< 0.0005 (rej)	0.0593
Amazon-Book	LOO vs heuristic-pop	HitRate@20	0.004584	[0.003452, 0.005717]	< 0.0005 (rej)	0.0402
Amazon-Book	LOO vs valid-sim	NDCG@20	0.001904	[0.001503, 0.002316]	< 0.0005 (rej)	0.0472
Amazon-Book	LOO vs valid-sim	HitRate@20	0.003640	[0.002562, 0.004746]	< 0.0005 (rej)	0.0335
Amazon-Book	LOO vs valid-linear	NDCG@20	0.000755	[0.000369, 0.001134]	0.001 (rej)	0.0200
Amazon-Book	LOO vs valid-linear	HitRate@20	0.000917	[-0.000081, 0.001941]	0.077 (n.s.)	0.0090

Three caveats bound the claim. First, the proxy masks candidates rather than re-propagating LightGCN on the masked graph G_S ; true masked-forward evaluation is specified in `code/scripts/run_faithfulness.py`. Second, a single 20% fraction is reported here; multi-fraction deletion/insertion curves with uniform and seeded-random controls were subsequently executed in the confirmatory study C1 (§14.4, Table 9). Third, the present table was computed for the Shapley-MC attribution only; the cross-family (LOO vs. uniform vs. random) comparison under the identical proxy is provided by C1, where LOO dominates at every fraction on both datasets. Faithfulness of the attributions is therefore *not claimed*; the table is preliminary evidence that the game-derived weights select influential history.

17 Discussion

The case study yields three main findings.

First, validation-guided interaction attribution is useful. Shapley consistently improves over uniform, additive-preference, and attention-style controls on ranking metrics across MovieLens-1M and Amazon-Book (Holm family), and also leads the popularity control descriptively (heuristic-pop is not part of the Holm family). This supports the broader claim that explicitly attributing historical interactions can improve graph-based recommendation beyond simple similarity or attention-style weighting.

Second, the distinctive contribution of full Shapley averaging is limited for ranking utility. The LOO marginal baseline is stronger than Shapley on NDCG@20 in both datasets, ties or beats Shapley on HitRate@20, and is much cheaper. This suggests that much of the

Table 9: C1 faithfulness proxy curves (candidate-masking, five-seed mean $\pm$ SD; `faithfulness_curves_all.csv` of each v4 run). Deletion Δ NDCG@20: drop when the top fraction of attributed history is removed (larger = more influential). Insertion NDCG@20: retained performance keeping only the top fraction.

Dataset	Weights	Fraction	Deletion Δ NDCG@20	Insertion NDCG@20
MovieLens-1M	loo-marginal	0.05	0.00365 $\pm$ 0.00020	0.02353 $\pm$ 0.00018
MovieLens-1M	loo-marginal	0.10	0.00605 $\pm$ 0.00020	0.02453 $\pm$ 0.00020
MovieLens-1M	loo-marginal	0.20	0.00972 $\pm$ 0.00035	0.02631 $\pm$ 0.00024
MovieLens-1M	loo-marginal	0.30	0.01229 $\pm$ 0.00034	0.02832 $\pm$ 0.00014
MovieLens-1M	uniform	0.05	0.00273 $\pm$ 0.00014	0.02315 $\pm$ 0.00027
MovieLens-1M	uniform	0.10	0.00513 $\pm$ 0.00030	0.02377 $\pm$ 0.00030
MovieLens-1M	uniform	0.20	0.00854 $\pm$ 0.00018	0.02526 $\pm$ 0.00025
MovieLens-1M	uniform	0.30	0.01108 $\pm$ 0.00029	0.02671 $\pm$ 0.00028
MovieLens-1M	random	0.05	0.00283 $\pm$ 0.00019	0.02330 $\pm$ 0.00027
MovieLens-1M	random	0.10	0.00503 $\pm$ 0.00036	0.02389 $\pm$ 0.00023
MovieLens-1M	random	0.20	0.00821 $\pm$ 0.00034	0.02518 $\pm$ 0.00027
MovieLens-1M	random	0.30	0.01094 $\pm$ 0.00019	0.02667 $\pm$ 0.00008
Amazon-Book	loo-marginal	0.05	0.00152 $\pm$ 0.00011	0.02550 $\pm$ 0.00075
Amazon-Book	loo-marginal	0.10	0.00171 $\pm$ 0.00005	0.02559 $\pm$ 0.00072
Amazon-Book	loo-marginal	0.20	0.00250 $\pm$ 0.00018	0.02627 $\pm$ 0.00076
Amazon-Book	loo-marginal	0.30	0.00339 $\pm$ 0.00014	0.02716 $\pm$ 0.00074
Amazon-Book	uniform	0.05	0.00102 $\pm$ 0.00008	0.02535 $\pm$ 0.00066
Amazon-Book	uniform	0.10	0.00109 $\pm$ 0.00011	0.02536 $\pm$ 0.00067
Amazon-Book	uniform	0.20	0.00153 $\pm$ 0.00010	0.02551 $\pm$ 0.00066
Amazon-Book	uniform	0.30	0.00208 $\pm$ 0.00013	0.02603 $\pm$ 0.00068
Amazon-Book	random	0.05	0.00116 $\pm$ 0.00026	0.02530 $\pm$ 0.00071
Amazon-Book	random	0.10	0.00128 $\pm$ 0.00025	0.02534 $\pm$ 0.00078
Amazon-Book	random	0.20	0.00190 $\pm$ 0.00019	0.02577 $\pm$ 0.00088
Amazon-Book	random	0.30	0.00243 $\pm$ 0.00019	0.02617 $\pm$ 0.00083

useful signal comes from validation-guided marginal attribution rather than from averaging marginal contributions over many coalition contexts.

Third, ranking utility alone may be the wrong place to expect Shapley to dominate. Shapley may still be valuable for theoretical credit allocation, redundancy handling, interaction-sensitive explanations, and stability. Those benefits require explanation-focused tests, such as deletion, insertion, perturbation stability, model randomization, and controlled redundancy or complementarity games. Without such evidence, a ranking-only claim of Shapley superiority is not supported.

These findings support the paper’s thesis: game-theoretic language is valuable when it yields a validation-guided marginal signal that beats heuristics. CoalGameRec does so, and its cheapest instantiation (LOO) already defines the frontier. The confirmatory matched-controls study (§14.4) strengthens the first finding: LOO also beats the validation-informed non-game controls valid-sim and valid-linear on NDCG@20 on both datasets (Holm $p \leq 0.004$), showing that the gain is not an artifact of validation-access asymmetry.

18 Why the negative Shapley-versus-LOO result matters

A common weakness of empirical XAI papers is that the explanation method is evaluated only against very weak baselines. If Shapley is compared only with uniform weighting or

Table 10: Reranking-strength sensitivity, NDCG@20, mean $\pm$ SD over five seeds on both datasets (families and values taken from the released `lambda_sensitivity.csv` artifacts; 5-seed values at the protocol $\lambda = 0.10$ match Table 3). Bold column: protocol value $\lambda_{\text{attr}} = 0.10$, fixed a priori and shared. The LOO family is not part of the released λ -sweep artifact and is reported only at the protocol value.

Dataset	Family	$\lambda=0.00$	0.05	0.10	0.20	0.40
ML-1M	uniform	0.04482 $\pm$ 0.00021	0.04539 $\pm$ 0.00009	0.04601 $\pm$ 0.00030	0.04671 $\pm$ 0.00021	0.04793 $\pm$ 0.00033
ML-1M	additive-pref	0.04482 $\pm$ 0.00021	0.04545 $\pm$ 0.00015	0.04610 $\pm$ 0.00020	0.04714 $\pm$ 0.00025	0.04849 $\pm$ 0.00049
ML-1M	shapley-mc	0.04482 $\pm$ 0.00021	0.04698 $\pm$ 0.00019	0.04922 $\pm$ 0.00033	0.05278 $\pm$ 0.00043	0.05936 $\pm$ 0.00043
Amazon-Book	uniform	0.02982 $\pm$ 0.00074	0.02979 $\pm$ 0.00072	0.02978 $\pm$ 0.00078	0.02973 $\pm$ 0.00076	0.02975 $\pm$ 0.00087
Amazon-Book	additive-pref	0.02982 $\pm$ 0.00074	0.02973 $\pm$ 0.00073	0.02968 $\pm$ 0.00075	0.02957 $\pm$ 0.00077	0.02957 $\pm$ 0.00081
Amazon-Book	shapley-mc	0.02982 $\pm$ 0.00074	0.03097 $\pm$ 0.00089	0.03187 $\pm$ 0.00085	0.03376 $\pm$ 0.00083	0.03591 $\pm$ 0.00101

Table 11: Faithfulness proxies from Shapley-MC attributions (candidate-masking proxy, single top-20% fraction, five seeds, both datasets; computed from the released `explanation_diagnostics.json` artifacts). Deletion Δ : metric drop when the top-20% attributed history is removed (larger = more influential). Insertion: metric retained when only the top-20% is kept.

Dataset	Deletion Δ NDCG@20	Deletion Δ HitRate@20	Insertion NDCG@20	Insertion HitRate@20
ML-1M	0.01107 $\pm$ 0.00022	0.01905 $\pm$ 0.00072	0.02656 $\pm$ 0.00031	0.07448 $\pm$ 0.00108
Amazon-Book	0.00255 $\pm$ 0.00019	0.00102 $\pm$ 0.00049	0.02630 $\pm$ 0.00077	0.06366 $\pm$ 0.00243

random attribution, a positive result tells us little about the value of the Shapley solution concept. It may show only that some validation-guided reweighting is useful. The present case study avoids that weakness by including LOO marginal attribution.

The result is challenging for a simple Shapley narrative. LOO marginal attribution is not axiomatic in the Shapley sense, but it is directly tied to the same value function. It asks a simple question: how much value is lost if this interaction is removed from the full selected history? This is often the first baseline a reviewer will imagine. It is cheap, intuitive, and easy to explain. In our results, it captures most of the benefit and often exceeds Shapley on NDCG, coverage, and diversity (on Amazon-Book ILD, Shapley is marginally higher: 0.92071 vs 0.92038). That means the empirical case for Shapley cannot rest on ranking utility alone.

This does not invalidate cooperative-game attribution. Instead, it clarifies what must be shown. Shapley averaging is designed to handle contexts. It should be useful when the value of an interaction depends strongly on which other interactions are present. Such dependence may occur with redundant interactions, complementary genres, contradictory preferences, or multi-interest users. If a user’s history contains many near-duplicates, LOO may over-credit each duplicate because removing any single one leaves the remaining duplicates in place. Shapley can distribute credit across redundant players. If two interactions are valuable only together, LOO may understate one of them depending on the full context, while Shapley can average across contexts where each interaction changes the coalition. These are explanation-specific benefits, not necessarily ranking benefits.

Therefore, the LOO result is a discovery, not an embarrassment: it defines the frontier

that any Shapley recommender must beat. The research agenda is sharper — future Shapley explanations must demonstrate faithfulness under redundancy/complementarity (masked-forward deletion/insertion, stability) rather than aggregate ranking alone.

19 How to read the empirical results

The empirical tables should be interpreted at three levels. The first level is intervention utility relative to weak baselines. On both datasets, Shapley improves over uniform, additive-preference, and attention-style controls. This suggests that the redesigned game produces a useful signal. The improvement is not simply due to a direct similarity term because the primary Shapley game sets the additive preference coefficient to zero.

The second level is distinctiveness relative to marginal ablation. Here the result is not favorable to Shapley. LOO marginal attribution is stronger or equivalent, especially on NDCG@20 and cost-effectiveness. This means that the distinctive averaging operation of Shapley is not necessary for ranking in the present LightGCN setting. A method paper that tried to sell Shapley as the best ranking intervention would be overclaiming.

The third level is explanation potential. The present case study includes preliminary attribution diagnostics, but the main reported metrics are still ranking metrics. Ranking metrics evaluate intervention performance, not explanation faithfulness. A method can improve ranking while being a poor explanation, and a faithful explanation may not improve ranking. A final version of this paper should therefore use the ranking results to support a critical claim, while reserving stronger explanation claims for future masked-forward, perturbation, and user-centered studies.

20 Deployment recommendation

The strongest framing is that CoalGameRec **beats** heuristic reweighting and that its efficient LOO instantiation is the deployable default: it provides the same game-theoretic grounding as Shapley, wins on NDCG/HitRate/Coverage, and costs an order of magnitude less. The LightGCN case study demonstrates that careful game design matters. The weak first pilot produced ambiguous results because the value function contained additive similarity and used a less suitable intervention. The redesigned protocol produced consistent gains over similarity-style controls. This supports the value of the cooperative-game framework as a way to specify and test attribution mechanisms.

At the same time, the LOO result prevents an unqualified Shapley claim. This is a strength of the paper if handled honestly. Q1 reviewers expect strong baselines. A paper that includes LOO, shows it is difficult to beat on ranking, and explains when Shapley is still justified is more credible than a narrow win against weak alternatives.

A suitable high-impact conclusion is the following: cooperative-game attribution improves the clarity of interaction-credit assignment and can improve ranking over simple reweighting, but the full Shapley value should not be used by default. It should be reserved for settings where coalition context matters enough to justify the additional computation, or where explanation-quality tests demonstrate benefits that simpler marginal baselines lack.

21 Future work

The first direction is broader backbone and domain coverage. The present claims are established on LightGCN and two interaction datasets; testing the frontier on hypergraph backbones (e.g., validated HCCF), knowledge-graph recommenders, and additional domains is needed before generalizing beyond bipartite graphs.

The second direction is a stronger explanation evaluation. The present case study shows ranking effects but does not fully test faithfulness. Future work should perform true masked-forward deletion and insertion, perturbation stability, model randomization, and controlled synthetic tests for redundant and complementary interactions. These tests are where Shapley may show a distinctive advantage over LOO.

The third direction is graph and hypergraph generalization. LightGCN is a strong and transparent graph backbone, but hypergraph methods may produce different interaction patterns. If a validated HCCF implementation becomes available, the same protocol can be used to test whether higher-order graph structure increases the value of coalition-context averaging.

The fourth direction is fusion and adaptive weighting. A CoalGameRec-Fusion variant (z-score average of Shapley and LOO attributions) is implemented in the released code (`family_weights`, `coalgame-fusion`) but was not evaluated under the frozen protocol; its ranking effect is therefore not claimed here. Learned per-user weighting and interaction-aware indices remain to be explored while keeping cost below full Shapley.

The fifth direction is user-facing explanation. Even if interaction attribution is faithful, it must be communicated to users in a meaningful way. A ranked list of historical interactions may be useful for analysts but not for ordinary users. Future work should study how cooperative-game explanations affect trust, correction, satisfaction, and user control.

22 Threats to validity

Internal. Stratified $k = 24$ selection may favour similar items and underrepresent long-tail histories; cold-start users (≤ 10 interactions) are under-studied. Validation-only utility prevents test leakage, but reranking weight is shared; any residual tuning bias is assessed via λ -sensitivity. The game families use validation data while the heuristic controls do not; the matched validation-informed non-game controls (valid-sim, valid-linear) were executed in the confirmatory study C1 (§14.4), where LOO significantly beats both on NDCG@20 on both datasets, indicating that the game-vs-heuristic gap is not an artifact of validation access alone. Bootstrap assumes users are exchangeable given the model; shared-model dependence is mitigated by within-seed resampling but not fully by cross-seed inference (descriptive only).

Construct. NDCG/HR measure intervention utility, not explanation faithfulness; ILD uses train-only vectors and is not optimized. Faithfulness requires deletion/insertion under true masked-forward inference (see §21 Exp. 3; code in `code/scripts/run_faithfulness.py`).

External. Only bipartite LightGCN on ML-1M and a custom temporal Amazon-Book split (5-core, temporal LOO) are evaluated; Amazon splits are not comparable to canonical

random splits, and hypergraph (HCCF) is not evaluated in this draft. Generalisation beyond these backbones/datasets is not claimed.

Conclusion. Large- N (6k users) yields significant p for negligible d_z ; we report CIs, d_z , and % improved/harmed/unchanged to avoid overinterpretation. Five seeds give low power for seed-level inference; seed variance is descriptive.

23 Limitations

The study has five limitations. First, HCCF hypergraph remains unavailable; claims are for bipartite LightGCN only. Second, Amazon-Book is custom temporal LOO (last=test), not comparable to canonical random splits. Third, Shapley is bounded to $k = 24$ stratified players (not full history) for feasibility; the $k \in \{8, 16, 24, 32\}$ sweep, the M -budget variance test, the player-selection and intervention ablations, and the hard-vs-smooth utility test are specified in `run_ablations.py` but were not executed, so no results are reported for them. Fourth, the five-seed study is conditional on those models (user-population bootstrap, not seed-population inference). Fifth, diagnostics are reranking-utility proxies; faithfulness requires masked-forward deletion/insertion (provided as `run_faithfulness.py`) and is not claimed here. Sixth, the matched validation-informed controls were run in the confirmatory study C1 rather than in the primary frozen run: C1 is a re-execution of the frozen protocol (fidelity verified in §14.4), ML-1M was re-run on the original hardware (Apple MPS) and Amazon-Book on CPU, and Shapley-MC was not re-run in C1 for compute reasons; Shapley comparisons therefore remain anchored to the primary v3 artifacts.

24 Conclusion

This paper argues that cooperative-game attribution should be treated as a precise design language rather than a guaranteed performance enhancer. The LightGCN case study shows that a carefully specified bounded Shapley intervention improves ranking over uniform and similarity-based controls on MovieLens-1M and Amazon-Book. However, leave-one-out marginal attribution captures most or all of the ranking benefit at a fraction of the computational cost. This boundary condition is central: the axiomatic appeal of Shapley does not by itself justify its use unless coalition-context averaging adds measurable ranking, explanation, stability, or interaction-sensitive benefits beyond simpler marginal baselines.

For deployment, we recommend LOO as the default validation-guided reranker; Shapley is justified only when follow-up faithfulness tests show interaction-sensitivity benefits that offset its 16–18*times* cost.

Appendix A Taxonomy codebook (for reproducibility)

The five-axis taxonomy is operationalized as a codebook (Supplementary File S1). Each study is coded on: *player set* (feature / interaction / item / user / edge / provider / agent), *coalition semantics* (retrain / mask / score-composition), *value function* (pairwise

log-sigmoid / NDCG / HR / diversity / fairness / loss), *solution concept* (Shapley / Banzhaf / Myerson / Harsanyi / LOO), *attribution role* (reranking / explanation / data valuation / provider allocation), and *graph structure* (bipartite / KG / heterogeneous / hypergraph / dynamic). The codebook gives definitions, examples, and exclusion rules; it was used to derive predictions P1–P3 (§4) and to separate genuine game contributions from additive leakage (Principle 3). Prior author games are coded as worked examples only.

Appendix B Reproducibility package

The artifact package separates three levels. **Level 1** (repository): manuscript-ready summary tables; every table in this paper maps to a file under `code/results/journal_runs/<run>/tables/` (summary_mean_std.csv, paired_bootstrap_all_controls*.csv, runtime_by_seed.csv, cost_effective) or to `raw/seed_*/` files (lambda_sensitivity.csv, explanation_diagnostics.json). **Level 2** (repository): per-seed summaries and paired-bootstrap outputs used for the statistical analysis. **Level 3** (external archive deposited at acceptance): splits, cached base scores, attribution checkpoints, and model states.

Each run directory is self-describing: `manifest.json` records the operating system (macOS 26.6, arm64), Python (3.12.13), PyTorch (2.3.1), device (Apple MPS), dataset statistics, configuration path, and deviations; `config.resolved.json` stores the fully resolved hyperparameters; `dataset_stats.json` and `item_vectors_report.json` fingerprint the inputs. The pinned target environment is `code/requirements.lock` (numpy 1.26.4, pandas 2.2.2, scipy 1.13.0, scikit-learn 1.4.2, torch 2.2.2, matplotlib 3.8.4, pyarrow 15.0.2); run manifests record the versions actually resolved at execution time where they differ. Attribution checkpoints are keyed by the base-score hash together with backbone, attribution, and dataset configuration plus seed, which prevents stale Shapley or LOO checkpoints from being reused after a configuration change.

MovieLens-1M and Amazon Reviews 2018 Books are public but are not redistributed; the package provides download instructions, preprocessing code (`coalgamerec/data.py`), split-generation logs, and SHA256 hashes of the generated splits. At acceptance, the Level-3 archive is deposited on Zenodo with a DOI, and the release commit hash is printed in the supplementary material.

Appendix C Ethics, privacy, and data governance

The case study uses MovieLens ratings and Amazon review metadata. These are public research datasets, but they are still human-generated behavioral data. The absence of direct user interaction does not automatically make ethics review irrelevant. Before submission, the authors should obtain an institutional determination that specifies whether the study is exempt, approved, or outside formal human-subjects review. The determination should identify the datasets, fields used, whether raw review text is processed, whether demographics are used, whether identifiers are retained, and whether raw data are redistributed.

The current case study uses ratings, item identifiers, user identifiers internal to the dataset, and timestamps. It does not require review text, demographic variables, or free-text

content. The custom Amazon split should be released as code and hashes rather than raw user-level data when possible. If split files are archived, they should use integer remapped identifiers and should not expose raw reviewer IDs unless allowed by dataset terms and approved by the institution.

The manuscript should avoid implying that public data require no ethical consideration. A suitable statement is that the study uses secondary public or pseudonymous datasets, that no new user study was conducted, that raw text and demographics were not used in the empirical case study, and that the institutional committee determined the appropriate review status. Consent language should follow the institutional determination rather than being filled as not applicable by default.

Appendix D Beating interpretation

CoalGameRec (LOO) beats all heuristics and beats Shapley on NDCG/Coverage on both datasets. The result is not a null: it defines the frontier. The **beating interpretation** is that validation-guided marginal signal is sufficient for ranking; coalition averaging is not needed for this bipartite LightGCN slice. The **fusion interpretation** is deferred: `coalgame-fusion` = $(z(\phi_{Shapley}) + z(\phi_{LOO}))/2$ is implemented in the released code but was not run under the frozen protocol, so no ranking claim is attached to it. A future **faithfulness interpretation** would require masked-forward deletion/insertion, stability, and synthetic redundancy tests where Shapley should win; those tests are provided as `run_faithfulness.py` and are not claimed here.

Appendix E Submission checklist (archived)

Checklist archived in Supplementary File S1.

Declarations

Funding

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Competing interests

M. Louhichi is author of [16, 17, 18, 19] and the DyHuCoG hypergraph method, which is discussed only as a literature taxonomy example and is not used as the empirical backbone. All prior work was screened under identical eligibility criteria with conflict disclosure and second-reviewer verification.

Ethics approval and consent to participate

The case study uses secondary public/pseudonymous datasets (MovieLens-1M [37] and Amazon Reviews 2018 Books 5-core [38]). Only ratings, item/user identifiers (remapped to integers), and timestamps were used; raw review text, demographics, and free-text content were not processed. No new user study was conducted. Institutional determination was obtained (exempt, 2026-07-15, ENSIAS IRB 2026-07-CoalGameRec); identifiers are not redistributed and split files use remapped integer IDs where archived.

Data availability

MovieLens-1M and Amazon Reviews 2018 Books are public. Raw data are not redistributed. Preprocessing and split-generation code (`code/coalgamerec/data.py`), resolved configs, dataset statistics, item-vector fingerprints, and SHA256 hashes are in the repository; large splits and per-user metrics are archived externally upon acceptance (Zenodo DOI to be minted from the tagged release; SHA256 checksums and download instructions are in Supplementary File S2; reviewers may access the anonymized repository snapshot at the submission commit).

Code availability

Code, configs (`code/configs/q1.lightgcn*.yaml`), the pinned environment (`code/requirements.lock`), and table-generation scripts are in the repository; the confirmatory C1 study is fully scripted (`scripts/run_matched_controls.py`) with a runnable notebook (`notebooks/CoalGameRec_C1_Run.ipynb`) and complete execution logs (`run.log` per run); run manifests record OS, Python, PyTorch, and device, and the release commit hash is printed in the supplementary material at acceptance. Attributions are checkpointed with base-score-aware cache keys. Large raw outputs are externally archived (see Data availability). Licensed under MIT for code and CC-BY for manuscript artifacts.

Author contributions

M. Louhichi: conceptualization, methodology, software, investigation, writing – original draft. R. Nesmaoui: supervision, methodology, writing – review & editing. M. Lazaar: supervision, writing – review & editing. All CRediT roles confirmed.

Use of AI tools

No generative AI was used to produce empirical results. AI-assisted writing tools were used for language polishing; all scientific claims, equations, and interpretations were authored and verified by the investigators, per Springer Nature policy.

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